谐振腔耦合波束扫描天线

天线频段60-64GHz（毫米波）

1.汽车雷达传感器

1. **AWR6443**：单芯片60GHz到64GHz汽车雷达传感器，集成了微控制器（MCU）和雷达加速器，用于汽车防撞系统、自适应巡航控制等。
2. **AWR6843**：单芯片60GHz到64GHz汽车雷达传感器，集成了数字信号处理器（DSP）、微控制器（MCU）和雷达加速器，适用于汽车防撞系统、自适应巡航控制等。
3. **AWR6843AOP**：单芯片60GHz到64GHz汽车雷达传感器，集成了天线封装（AoP）、数字信号处理器（DSP）和微控制器（MCU），用于汽车防撞系统、自适应巡航控制等。

2.工业毫米波雷达传感器

1. **IWR6443**：单芯片60GHz到64GHz工业毫米波传感器，集成了微控制器（MCU）和硬件加速器，用于工业自动化、机器人定位等。
2. **IWR6843**：单芯片60GHz到64GHz工业毫米波传感器，集成了处理能力，适用于工业自动化、机器人定位等。
3. **IWR6843AOP**：单芯片60GHz到64GHz工业毫米波传感器，集成了天线封装（AoP），用于工业自动化、机器人定位等。

3.无线通信设备

1. **IWRL6432**：单芯片低功耗57GHz到64GHz工业毫米波雷达传感器，适用于短距离高速数据传输。
2. **IWRL6844**：单芯片低功耗57GHz到64GHz工业毫米波雷达传感器，适用于短距离高速数据传输。

4.安防监控设备

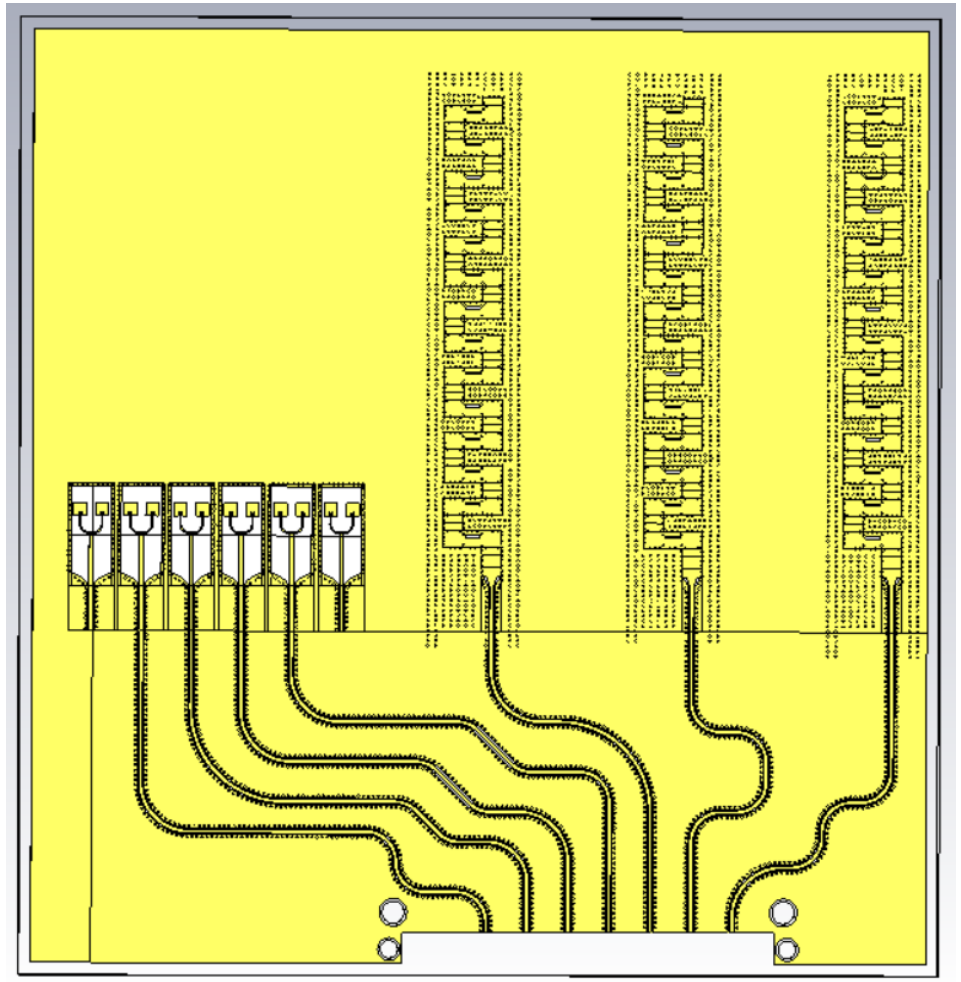
1. 毫米波成像设备：利用60GHz到64GHz频段的毫米波进行成像，用于安检、监控等领域，能够穿透衣物等障碍物，检测隐藏的物体。

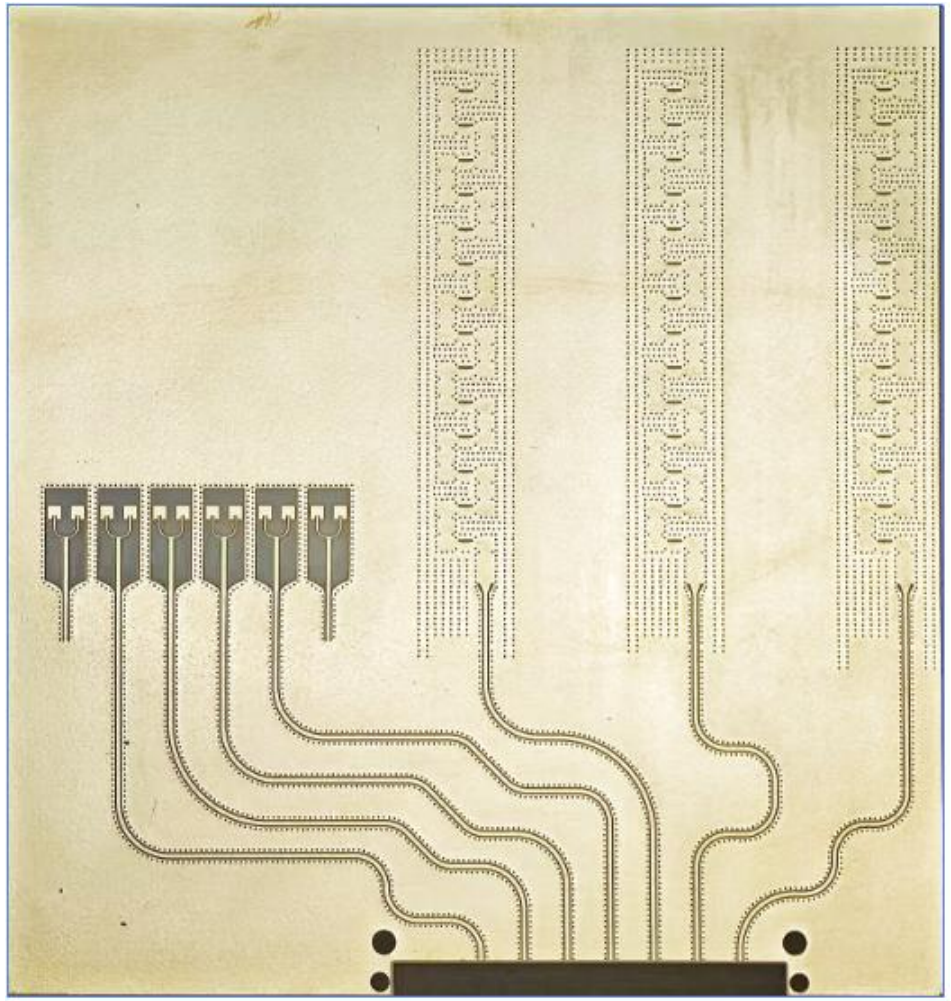
5.医疗设备

1. 毫米波治疗设备：利用毫米波的生物效应，用于医疗治疗，如促进血液循环、缓解疼痛等。

"E:\课题组项目\天线设计\三阶最佳\last-1.5 - array - 2.cst"

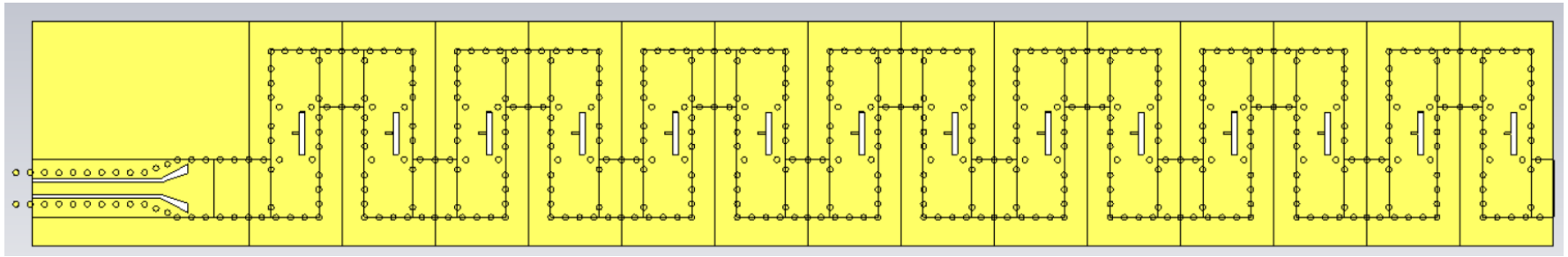
整体结构与加工图片（三发四收）



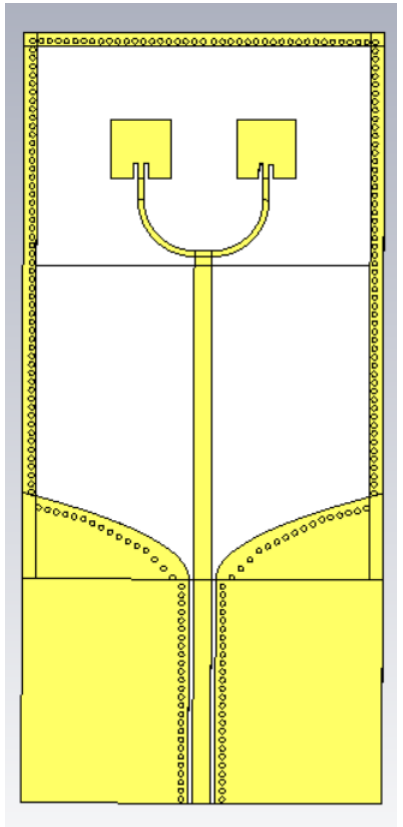


三个模块组件

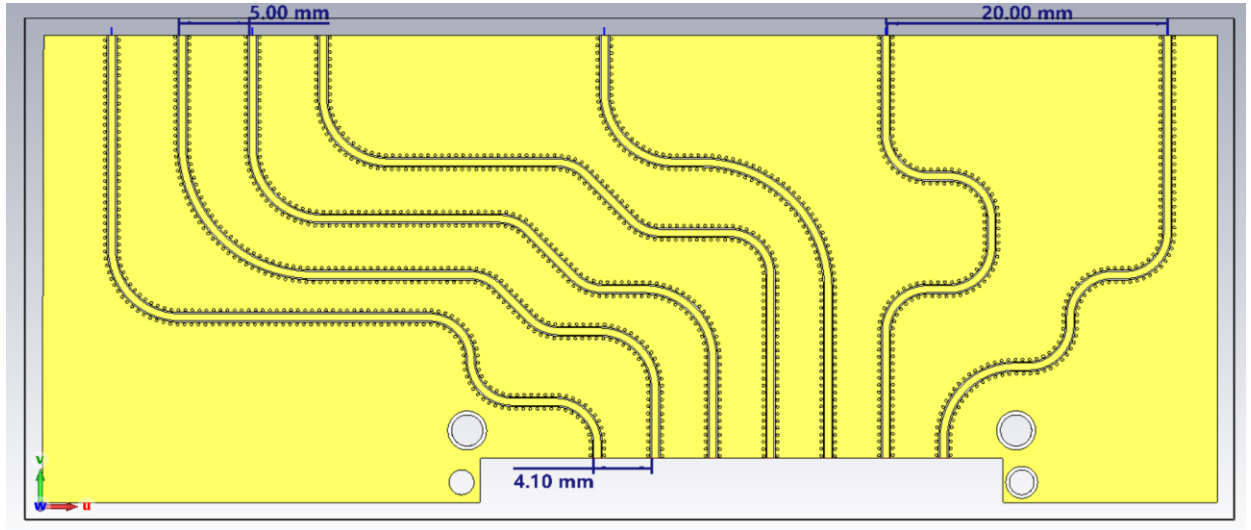
发射天线



接收天线



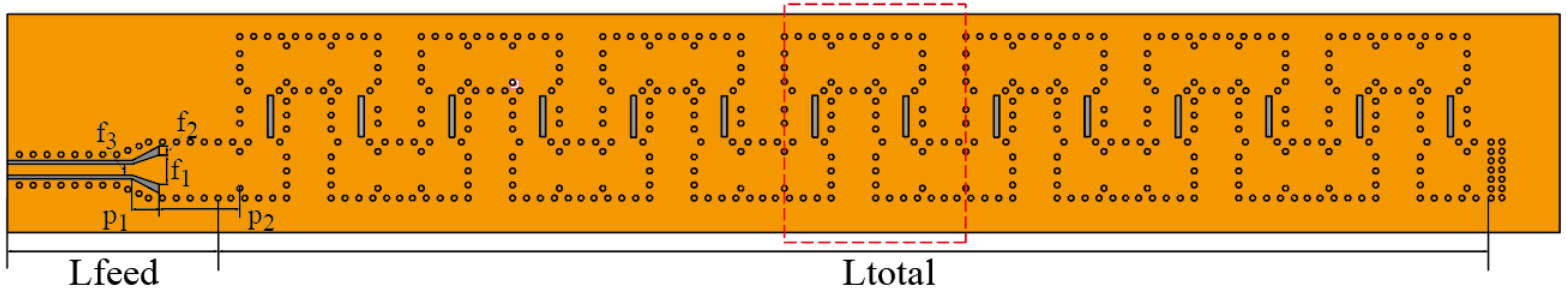
转接器



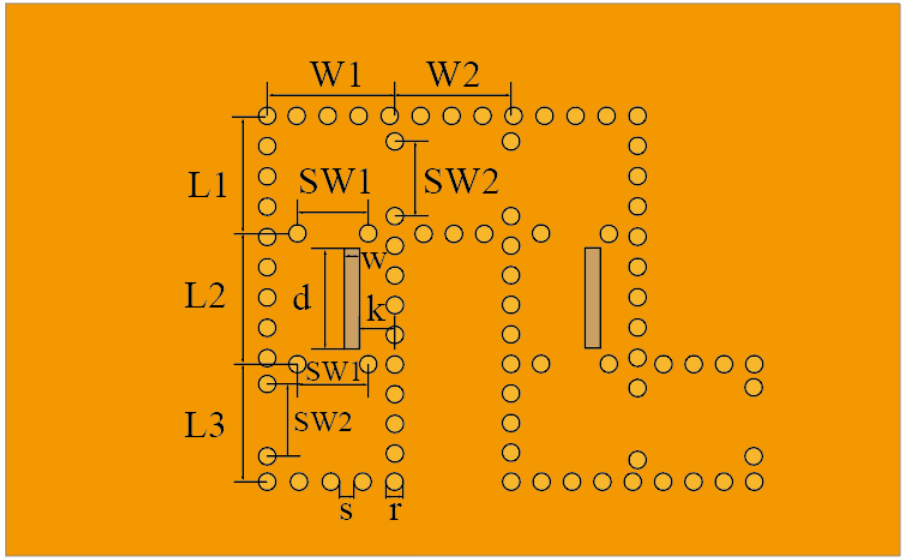
材料参数

介质基板：厚度0.254mm(10mil)　材料RO3003（$\varepsilon_{\mathrm{r}} = 3, \delta = 0.00$ ）
铜厚：　1oz（35um）　　表面处理方式：沉金
过孔（孔内镀铜）　孔径 0.2mm(直径)　　孔壁间距0.3mm

"E:\课题组项目\天线设计\三阶最佳\3阶 - 最终1.cst"



(a)



(b)

Parameter	*W1*	*W2*	*L1*	*L2*	*L3*	*w*
Value (mm)	1.7	1.56	2	1.84	2	0.1
Parameter	*SW1*	*SW2*	*d*	L_{total}	L_{feed}	*k*
Value (mm)	1.1	1.3	1.46	45.64	7.58	0.6
Parameter	*P1*	*P2*	*f1*	*f2*	*f3*	
Value (mm)	0.96	2.9	1.05	0.32	0.4	

先导结果2阶

扫描角度不够

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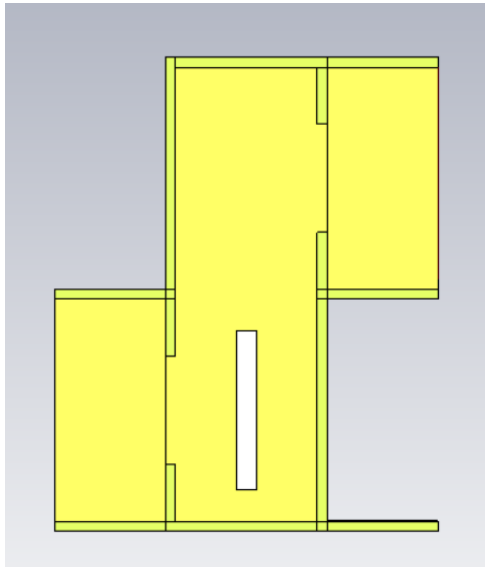
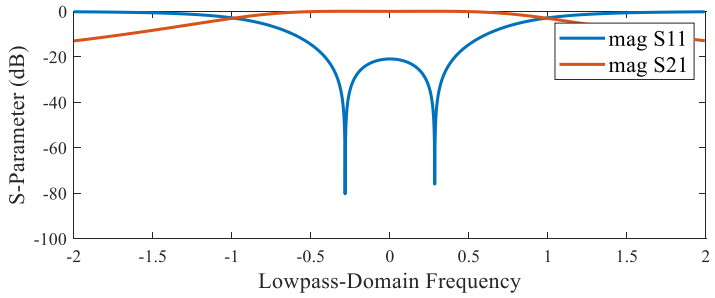
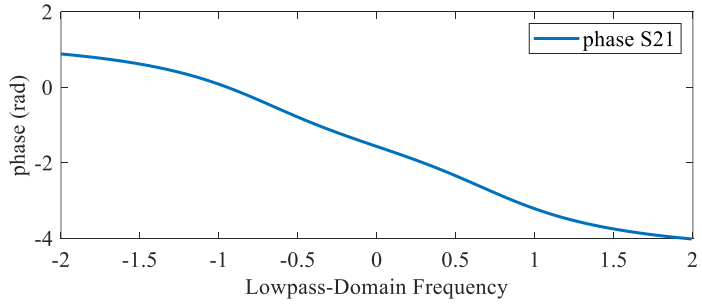
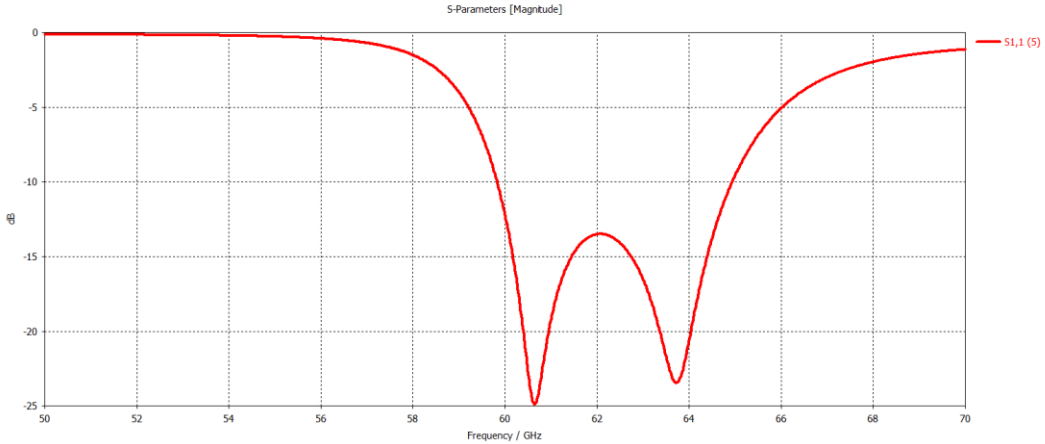


matlab模拟结果

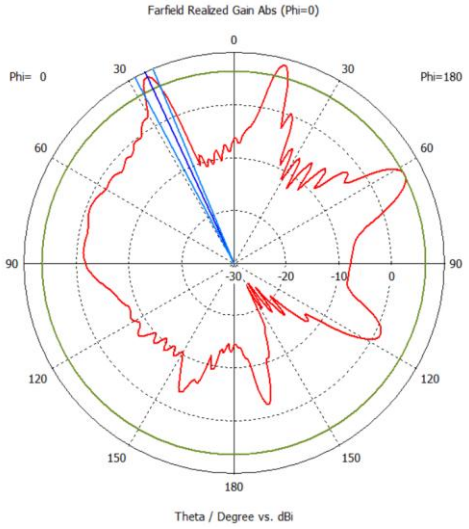


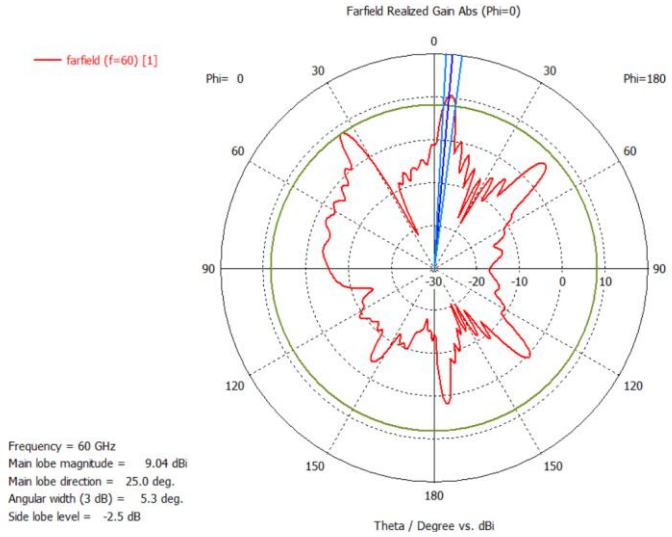


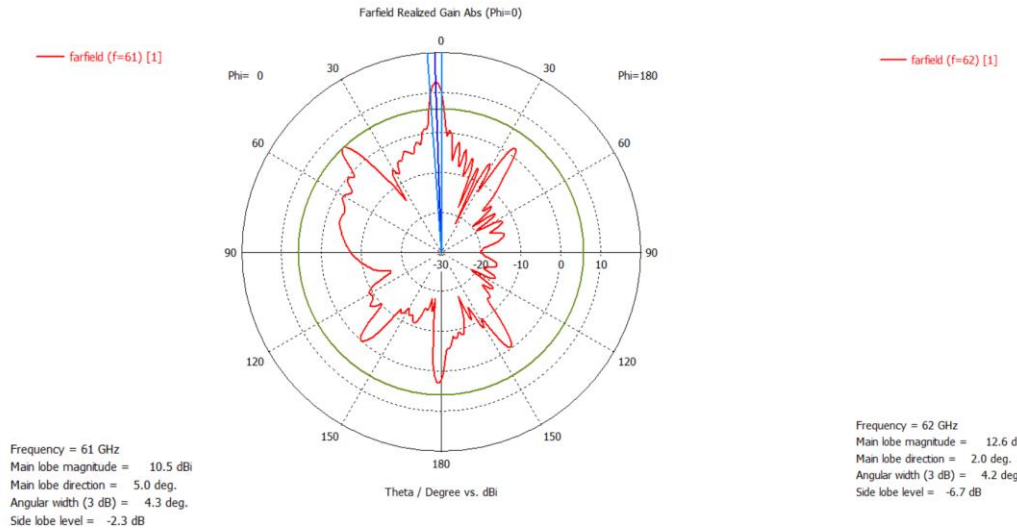
CST仿真结果

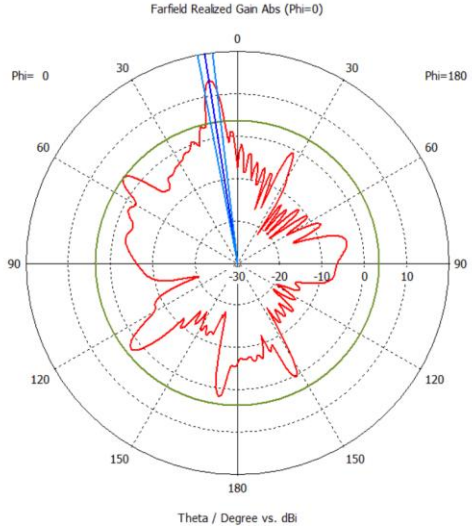


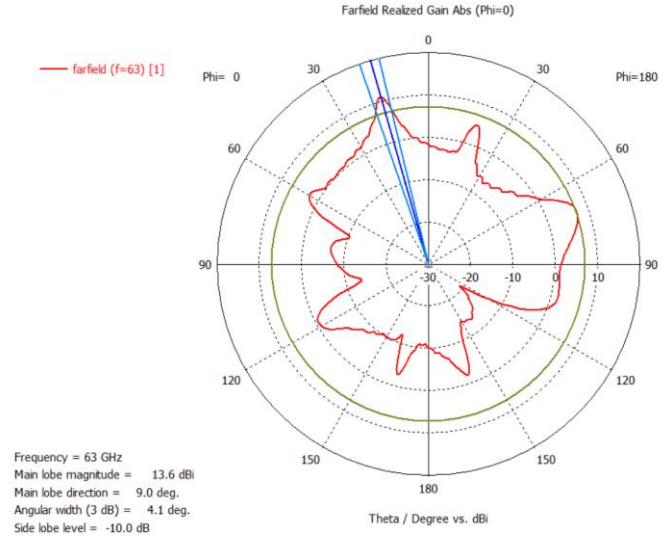
先导结果2阶　　扫描角度不够

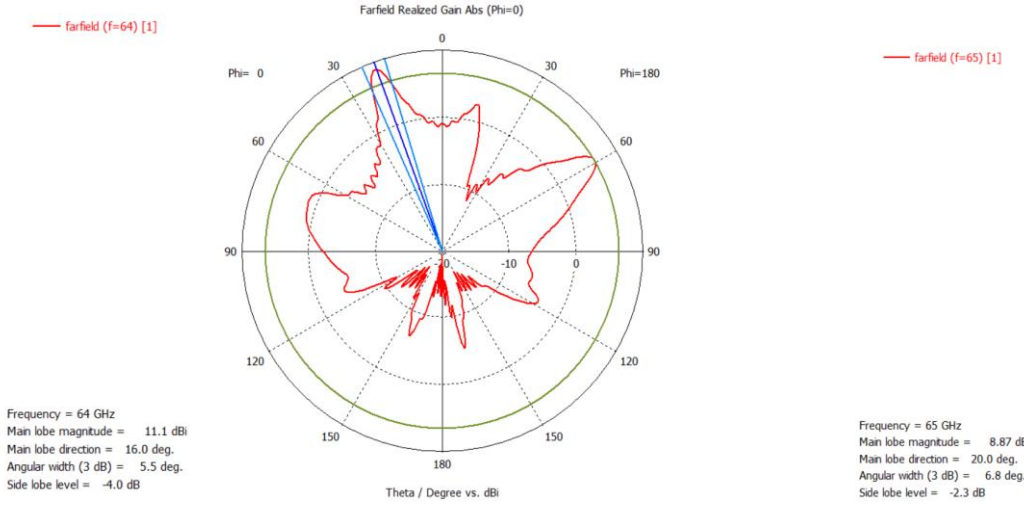






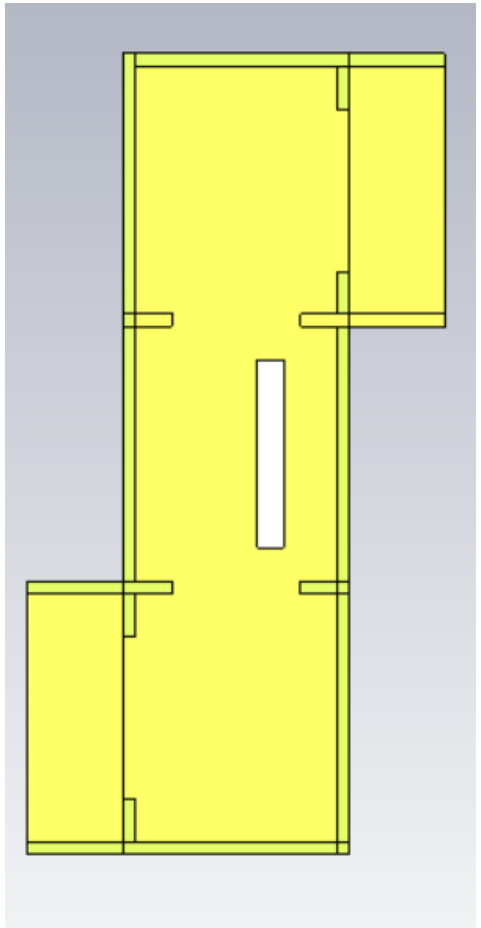




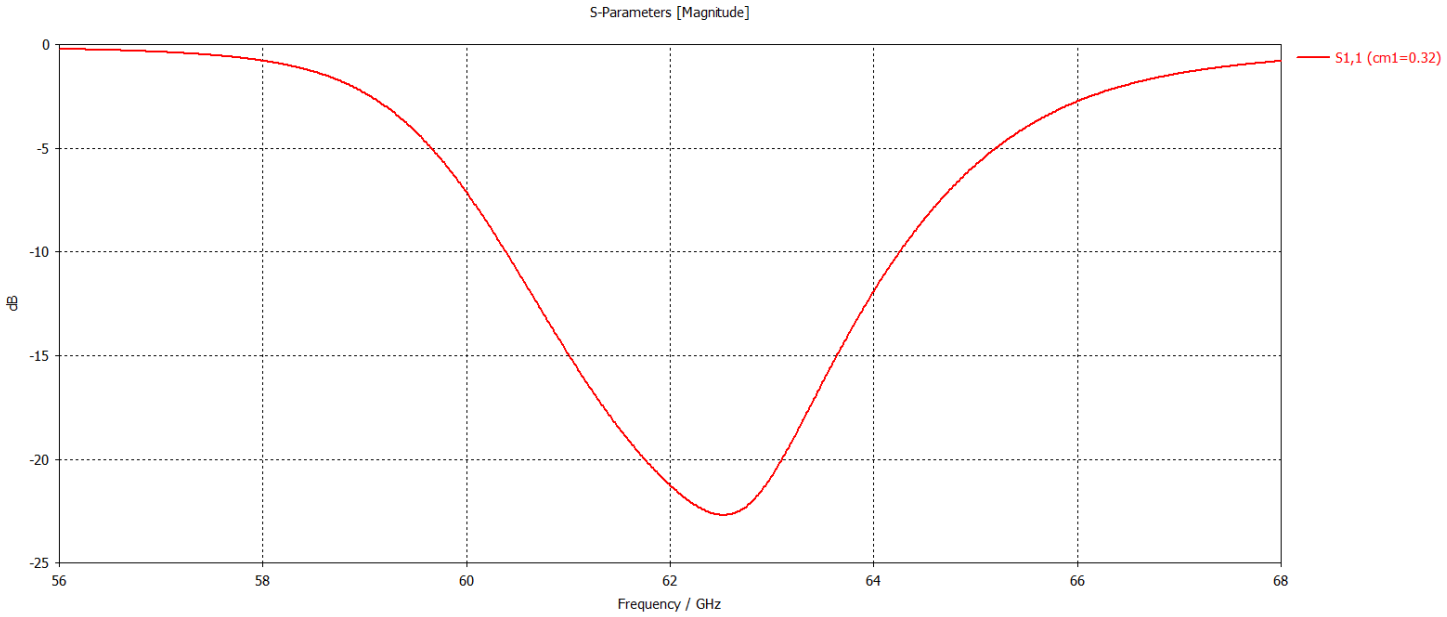


单元仿真

"E:\课题组项目\天线设计\波束扫描天线设计项目\腔体天线3阶\三阶竖缝调参 - 3 - unitwell.cst"



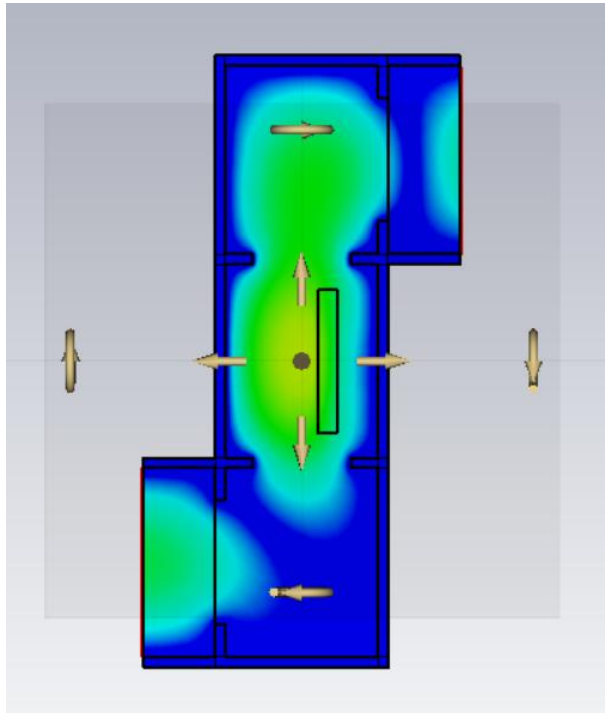
单元为三个谐振腔耦合，中间谐振腔开缝进行辐射，辐射缝隙偏移是为了耦合强度的调节和电磁场的分布，耦合方式为腔间耦合，包括电耦合、磁耦合和绕射耦合。单元之间使用波导连接，等效RLC电路见后。



确定各项参数方法

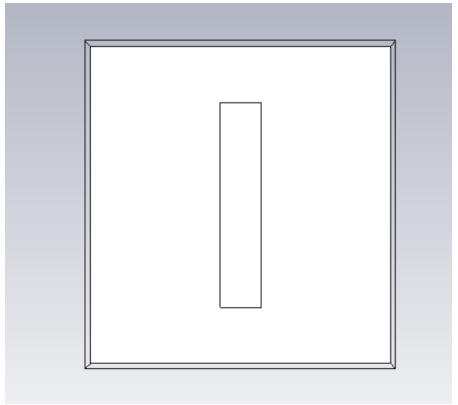
谐振频率调节

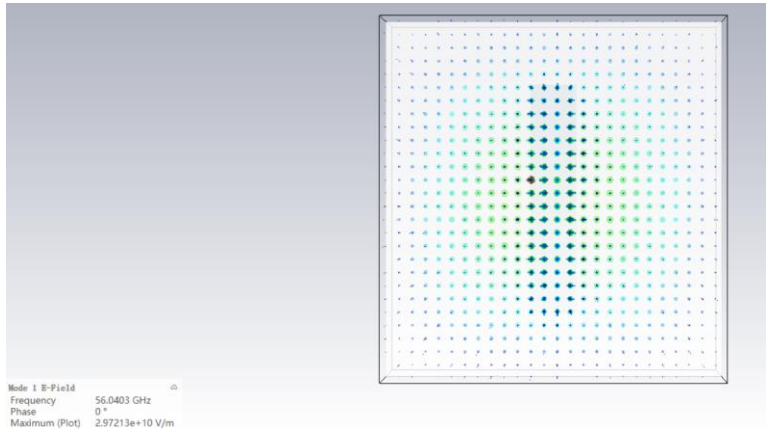
方法1——观察不同频率电场



E:\课题组项目\天线设计\T型三阶耦合波束扫描漏播天线\squre-本征模仿真分析

方法2——本征模





耦合调节与角度粗浅预测

耦合矩阵M是(N+1)×(N+1)的实对称矩阵：
主对角线：谐振腔自谐振频率（通常归一化为0）。
非对角线M(i,j)：谐振腔i与j之间的耦合强度。
输入端口（第1行/列）和输出端口（第N+1行/列）通过耦合系数与谐振腔连接。
通过调整物理尺寸（如缝隙宽度、谐振腔间距）实现耦合系数。

- **正耦合**：电容耦合或电场耦合。
- **负耦合**：电感耦合或磁场耦合。
- **零元素**：无直接耦合。

1. **归一化低通频率**：

$$w = \frac{W_0}{BW}\left(\frac{W}{W_0} - \frac{W_0}{W}\right)$$

   - $W_0 = \sqrt{f_1 f_2} \cdot 2\pi$（中心频率）
   - $BW = (f_2 - f_1) \cdot 2\pi$（带宽）

2. **阻抗矩阵**：

$$Z = w \cdot I - j \cdot R + M$$

   - I：单位矩阵
   - R：输入/输出端口阻抗矩阵
   - M：耦合矩阵

3. **S参数**：
   - 反射系数 S_{11}：

$$S_{11} = -1 - 2j \cdot R_1 \cdot Z_t(1,1)$$

   - 传输系数 S_{21}：

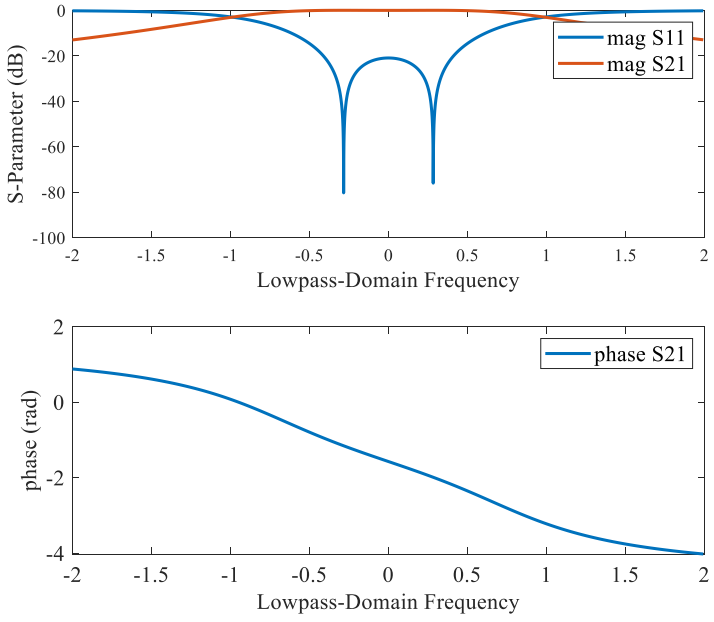
$$S_{21} = -2j \cdot R_1 \cdot Z_t(N+1,1)$$

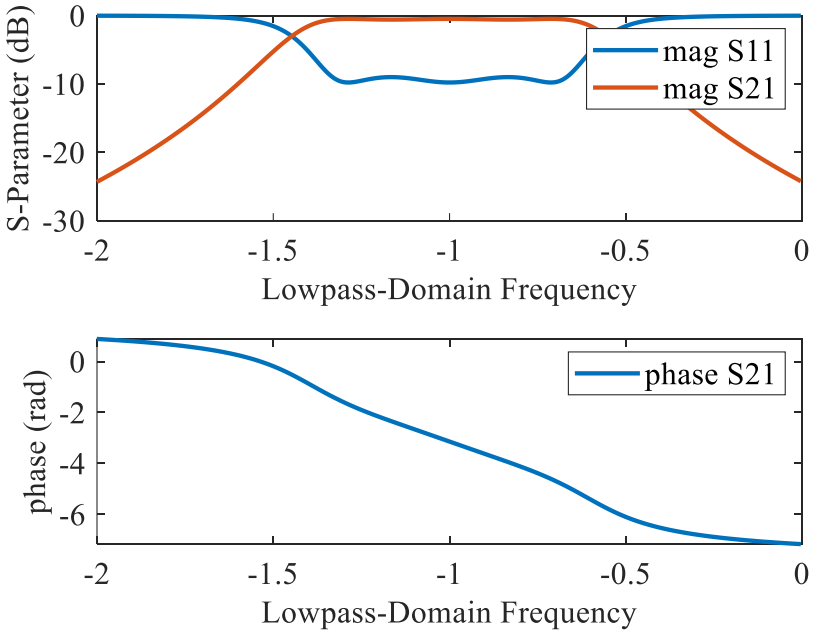
   - $Z_t = Z^{-1}$（阻抗矩阵的逆）

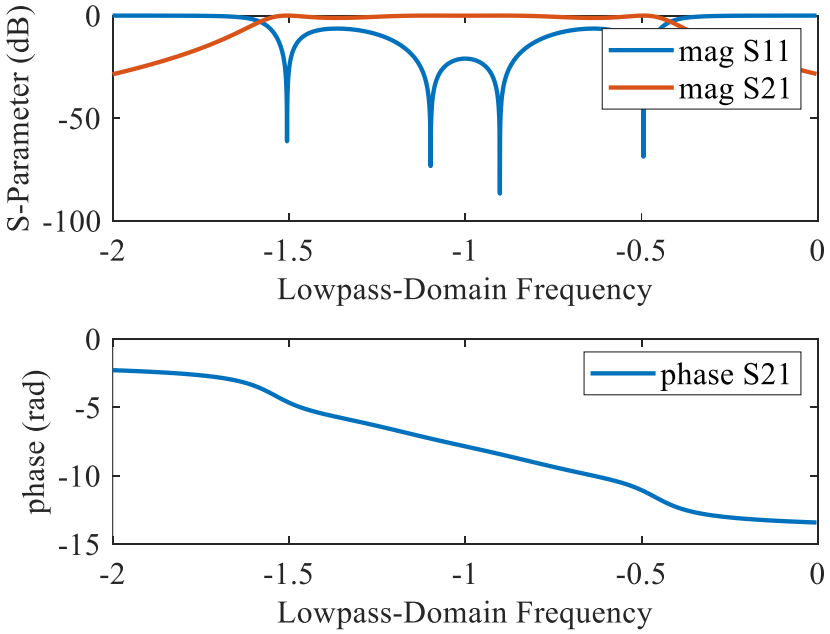
确定各项参数方法

耦合调节与角度粗浅预测

E:\课题组项目\matlab算法\耦合矩阵图像预测\4-t







扫描角度预测——使用S21（精确预测）

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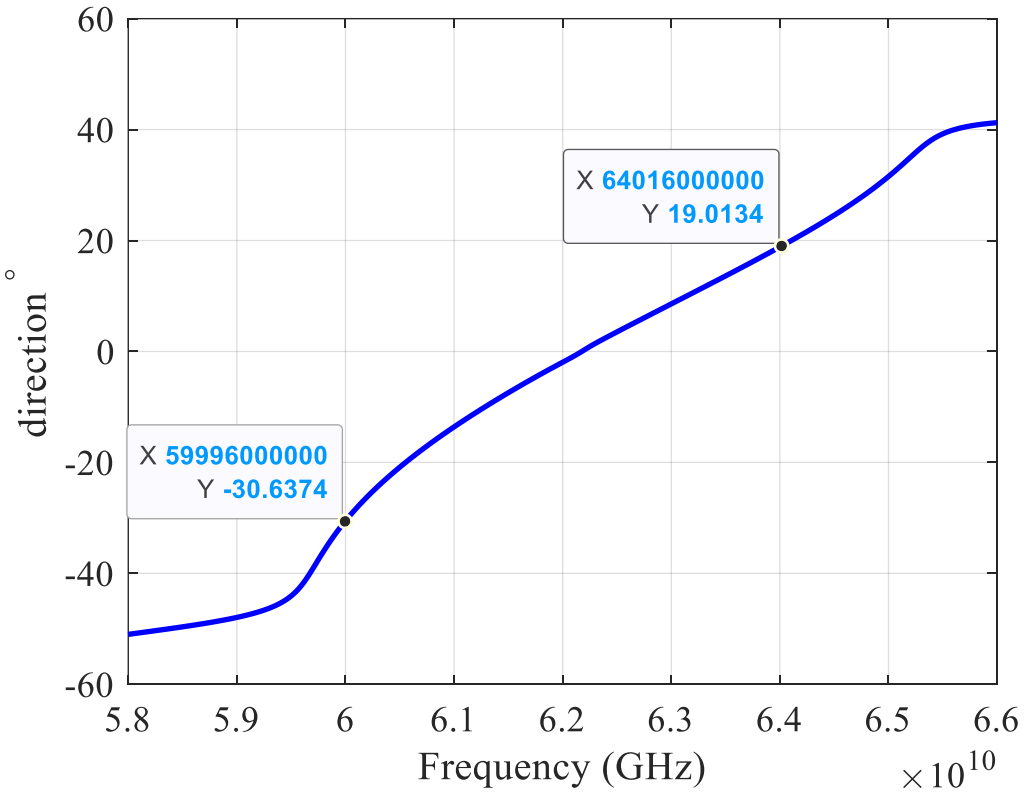


1. **AD参数**:

$$AD = \frac{1 - S_{22}S_{11} + S_{12}S_{21}}{S_{21}}$$

2. **传播常数**:

$$\gamma = \frac{\cosh^{-1}(AD/2)}{d}$$

3. **衰减常数**:

$$\alpha = \frac{\mathrm{Re}(\gamma)}{k_0}$$

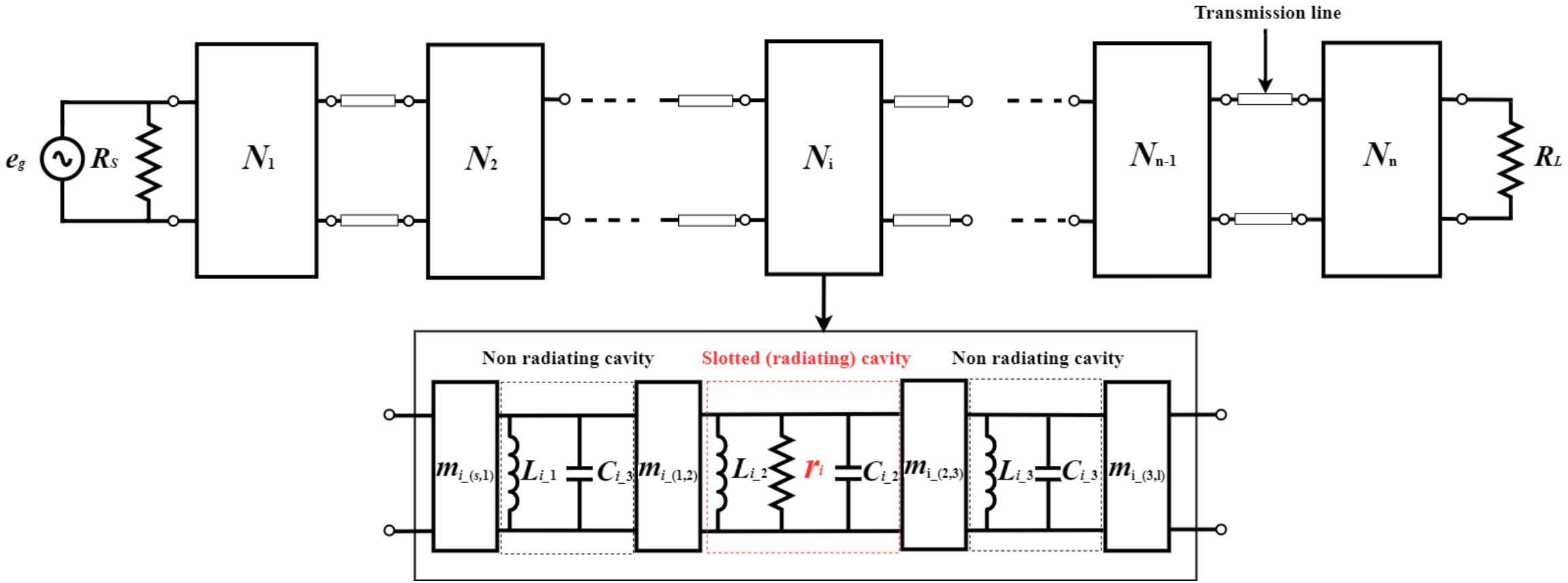
4. **相位常数**:

$$\beta = \frac{\mathrm{Im}(\gamma)}{k_0}$$

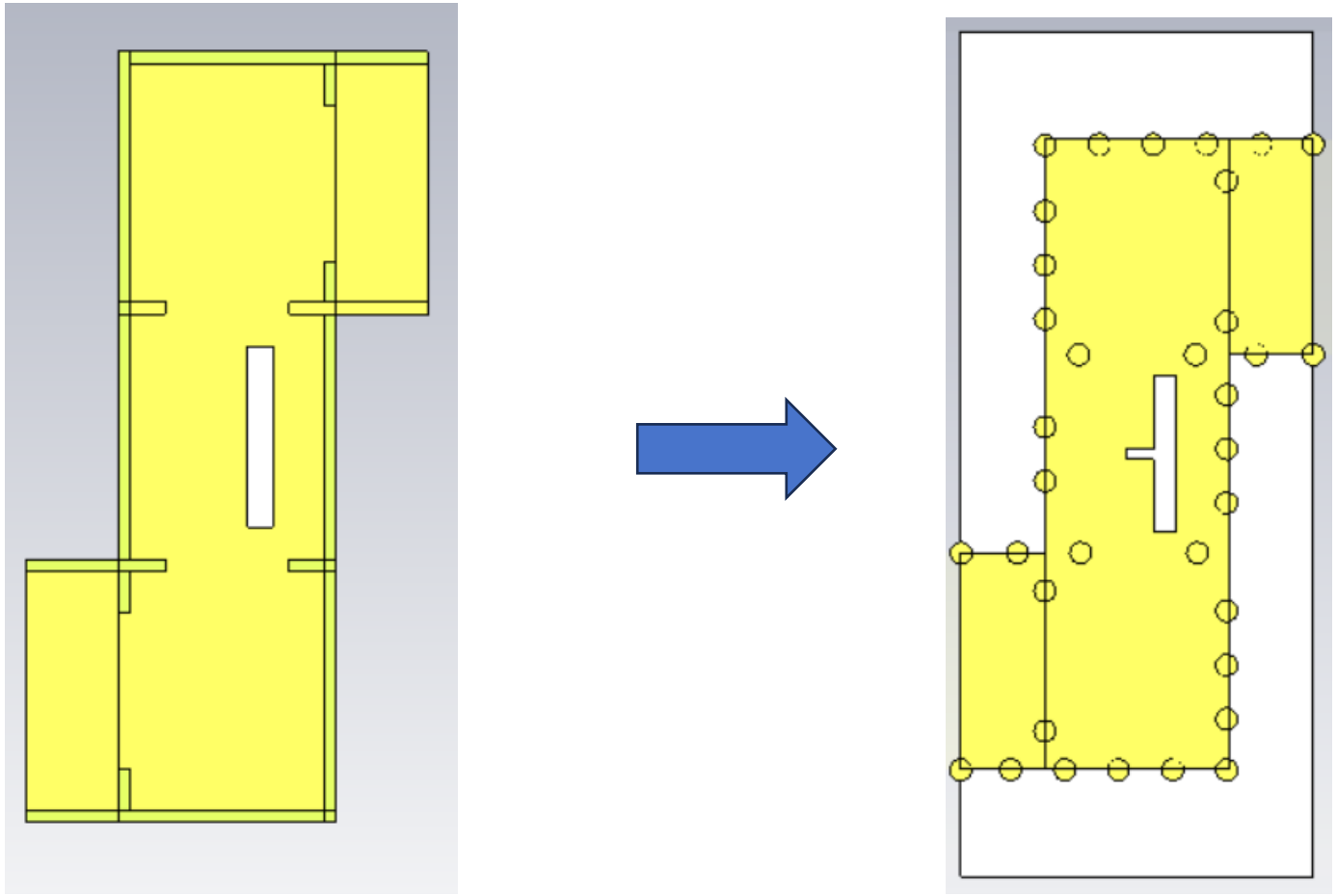
5. **特性阻抗**:

$$Z_b = \frac{-2BZ_0}{A - D + \sqrt{(A+D)^2 - 4}}$$

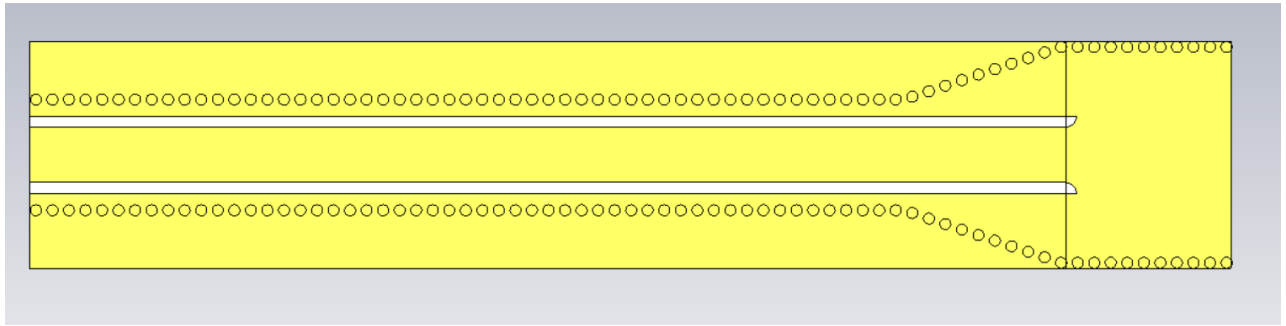
谐振腔耦合天线阵列的RLC图

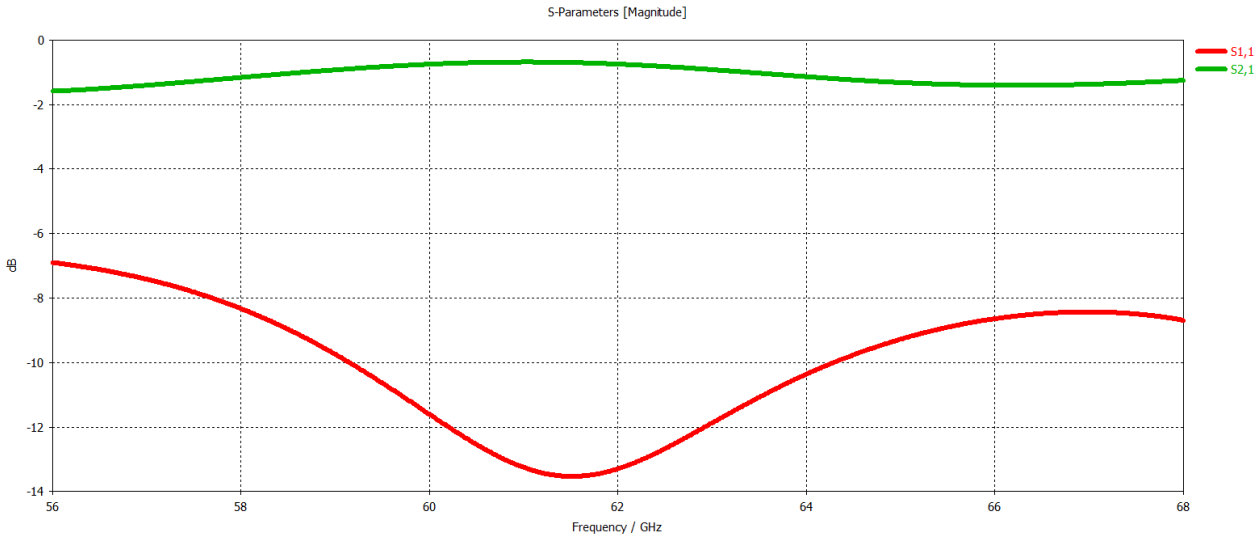


改造——OSB（开组带 去除结构的对称性）电壁转SIW

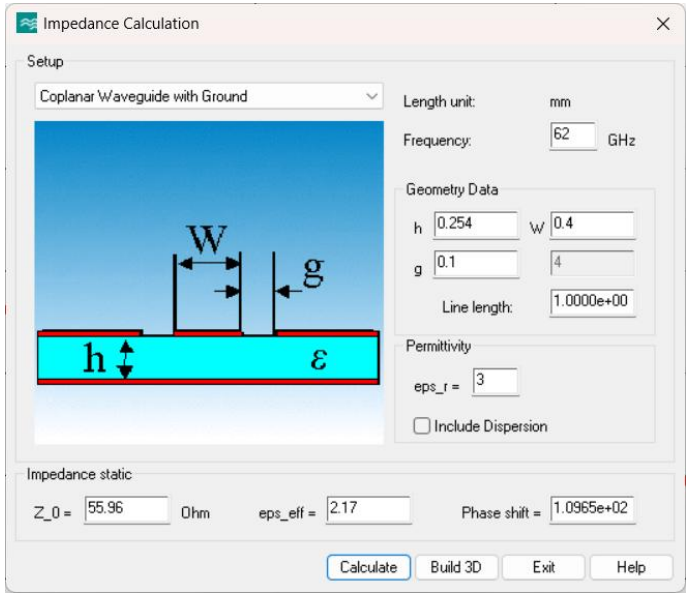


改造——GCPW转波导

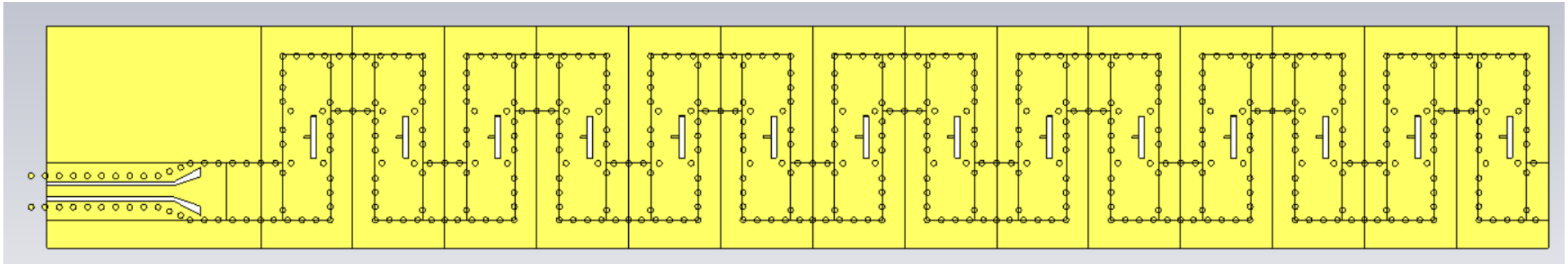


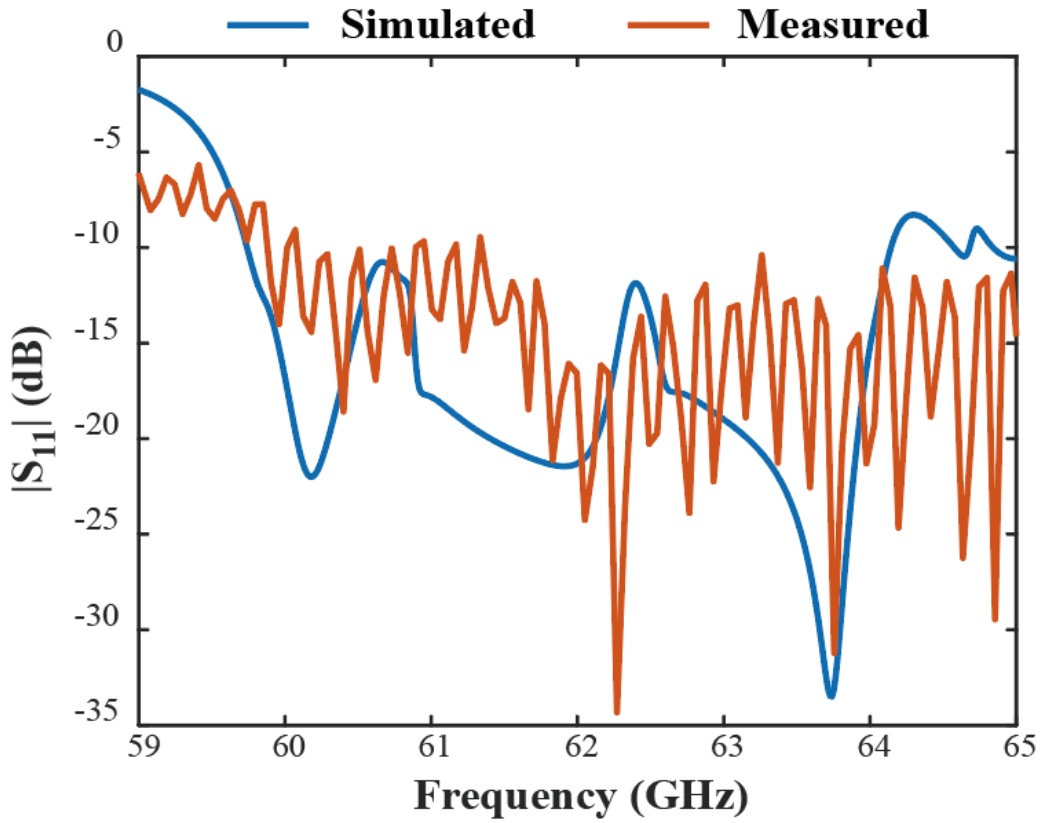


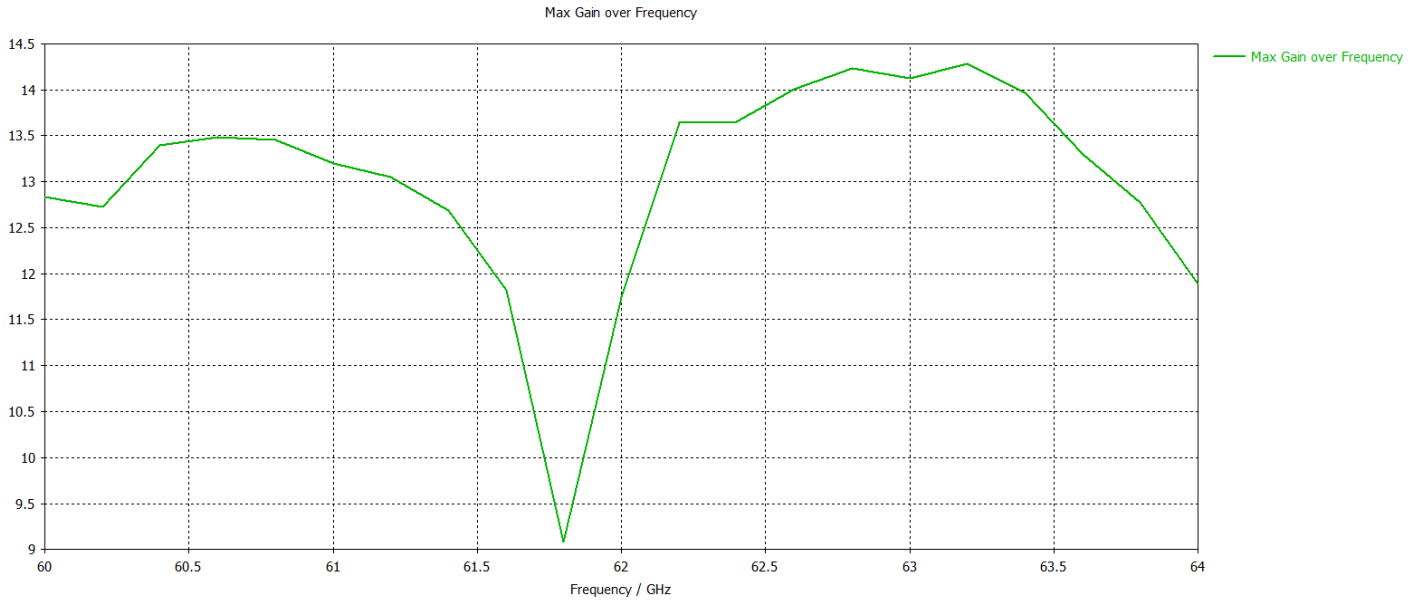
GCPW 参数 ：



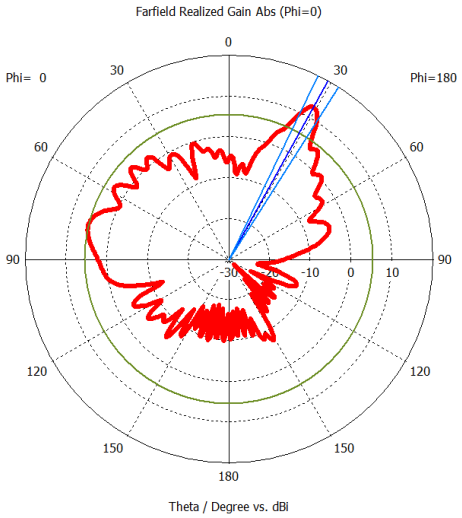
阵列化天线

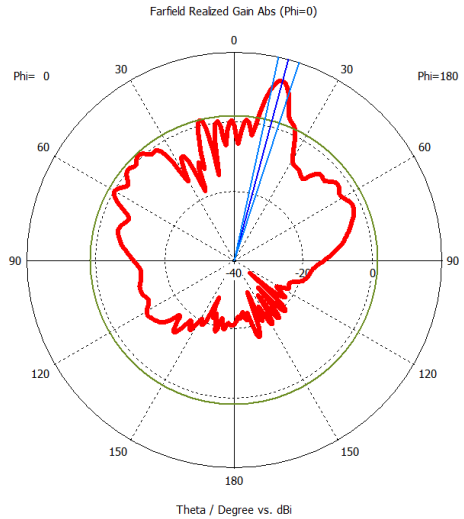


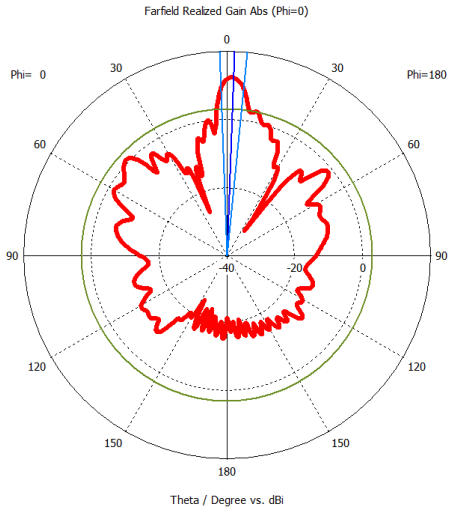

Simulated
Measured
0
-5
-10
-15
-20
-25
-30
-35
|S11| (dB)
59
60
61
62
63
64
65
Frequency (GHz)


Max Gain over Frequency
14.5
14
13.5
13
12.5
12
11.5
11
10.5
10
9.5
9
60
60.5
61
61.5
62
62.5
63
63.5
64
Frequency / GHz
Max Gain over Frequency

最终1







farfield (f=60) [1]

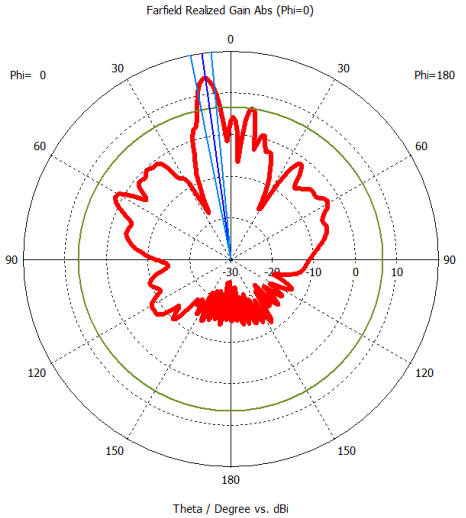
Frequency = 60 GHz
Main lobe magnitude = 12.6 dBi
Main lobe direction = 29.0 deg.
Angular width (3 dB) = 6.6 deg.
Side lobe level = -7.2 dB

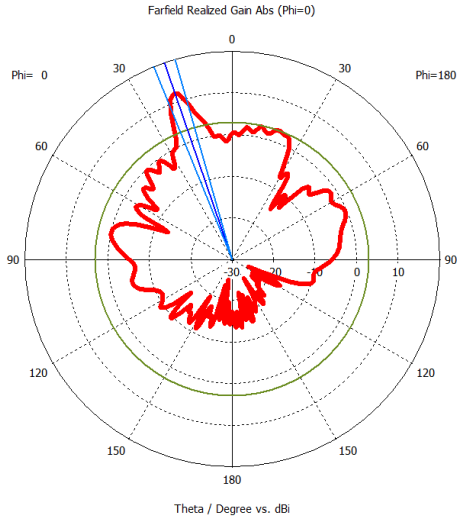
farfield (f=61) [1]

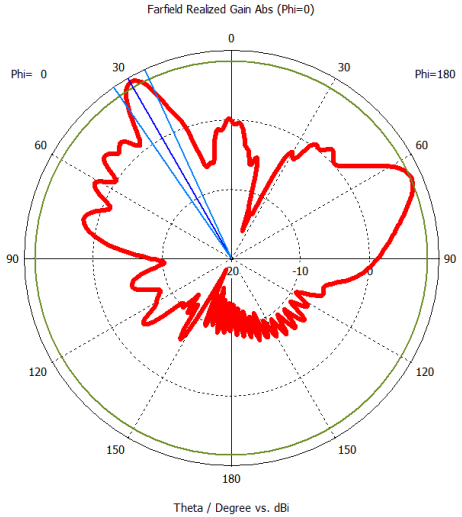
Frequency = 61 GHz
Main lobe magnitude = 13.5 dBi
Main lobe direction = 15.0 deg.
Angular width (3 dB) = 5.9 deg.
Side lobe level = -11.8 dB

farfield (f=62) [1]

Frequency = 62 GHz
Main lobe magnitude = 12.3 dBi
Main lobe direction = 2.0 deg.
Angular width (3 dB) = 7.7 deg.
Side lobe level = -9.2 dB







farfield (f=63) [1]

Frequency = 63 GHz
Main lobe magnitude = 14 dBi
Main lobe direction = 8.0 deg.
Angular width (3 dB) = 5.6 deg.
Side lobe level = -7.3 dB

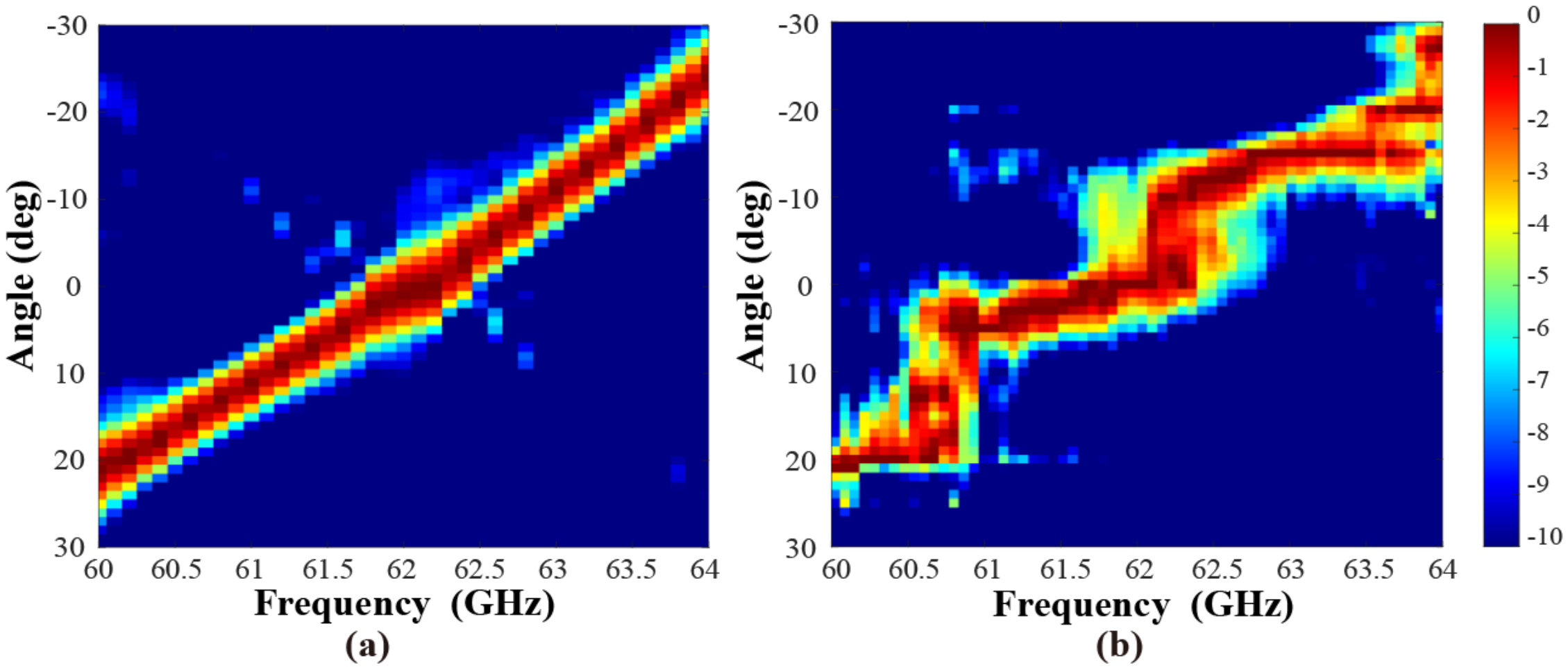
farfield (f=64) [1]

Frequency = 64 GHz
Main lobe magnitude = 12.2 dBi
Main lobe direction = 19.0 deg.
Angular width (3 dB) = 6.2 deg.
Side lobe level = -9.3 dB

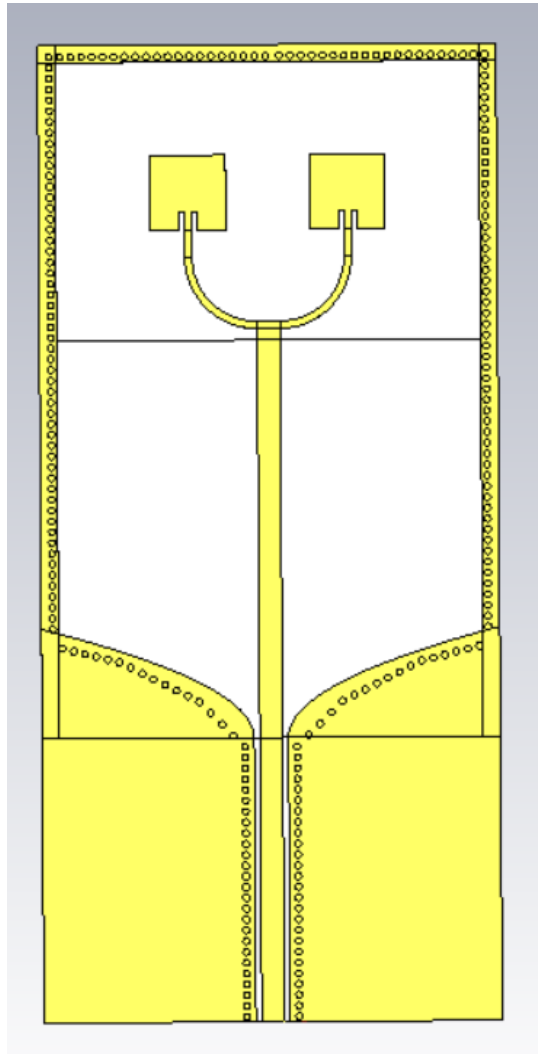
farfield (f=65) [1]

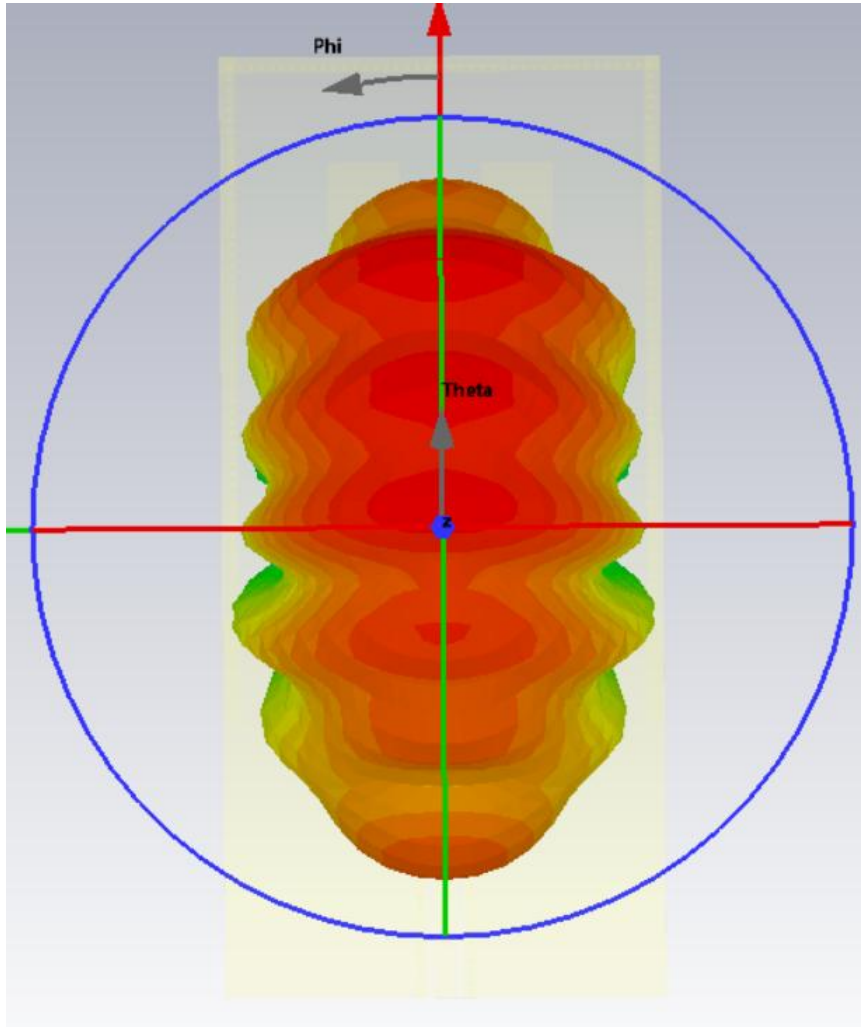
Frequency = 65 GHz
Main lobe magnitude = 9.26 dBi
Main lobe direction = 30.0 deg.
Angular width (3 dB) = 9.8 deg.
Side lobe level = -0.8 dB

加工后方向测试结果


-30
-20
-10
0
10
20
30
Angle (deg)
60
60.5
61
61.5
62
62.5
63
63.5
64
Frequency (GHz)
(a)
-30
-20
-10
0
10
20
30
Angle (deg)
60
60.5
61
61.5
62
62.5
63
63.5
64
Frequency (GHz)
(b)
0
-1
-2
-3
-4
-5
-6
-7
-8
-9
-10

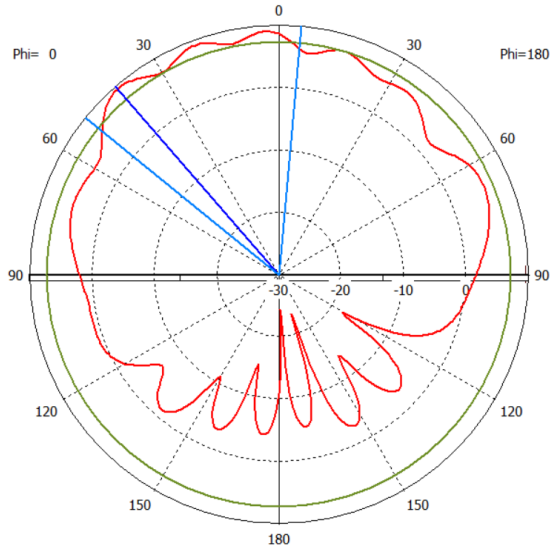
"E:\课题组项目\天线设计\波束扫描天线设计项目\微带接收天线\双微带 - Gcpw - 最终模式 - u - 2.cst"





要求：
极化方向相同：水平极化。
远场方向：纵向宽横向足够窄。

结构：
使用双微带天线压缩横向波束宽度（结构简单实用）。

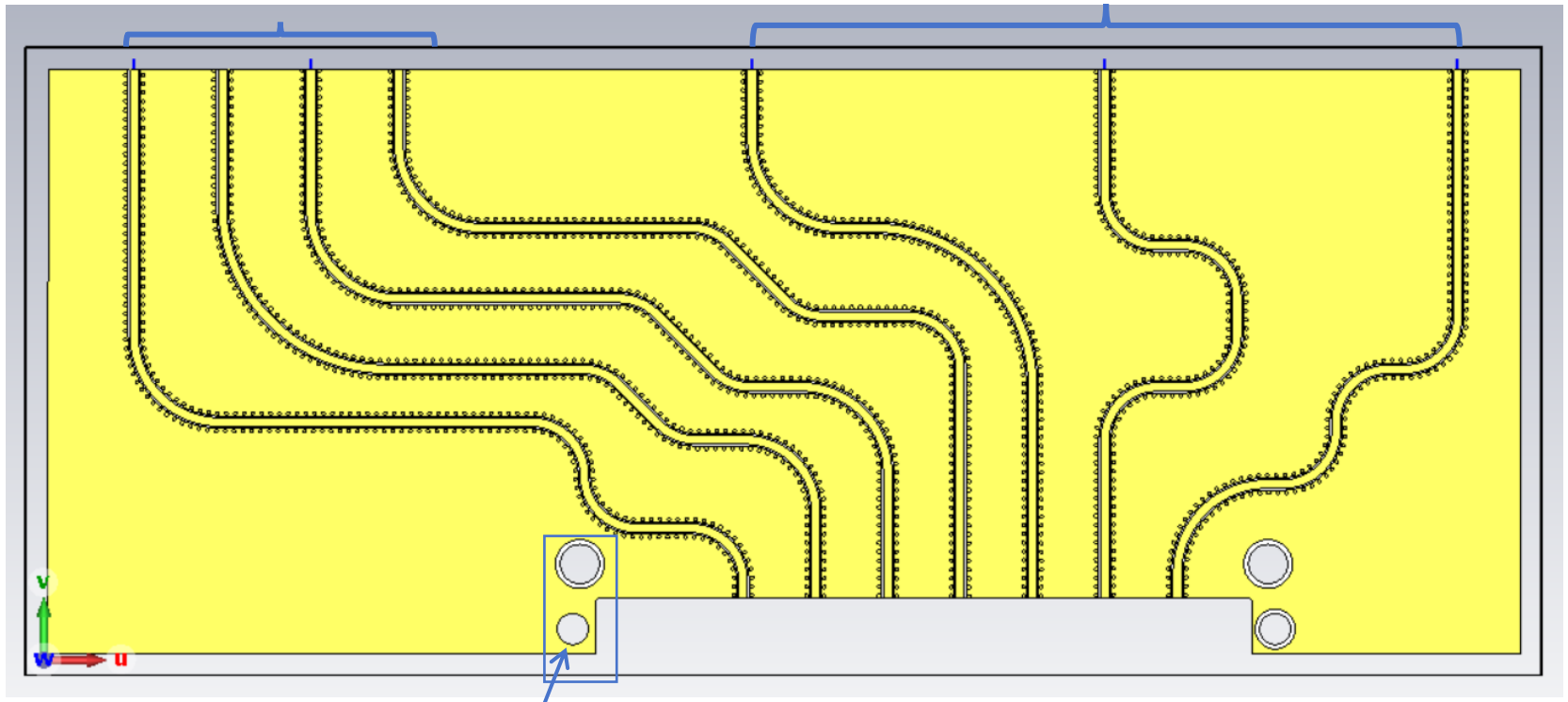


等相位转接线

- "E:\课题组项目\天线设计\三阶最佳\转接器-last-1.5.cst"

接收天线（间隔一倍波长）

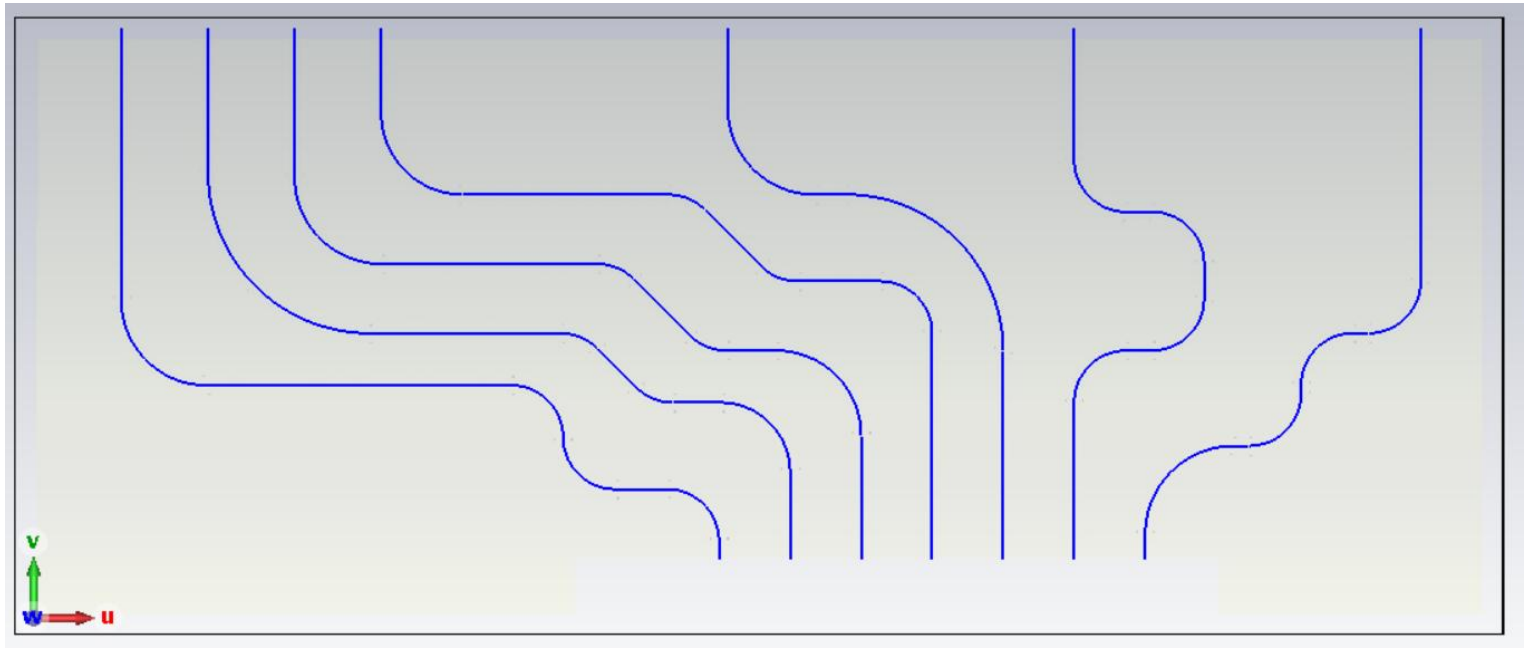
发射天线（间隔四倍波长）



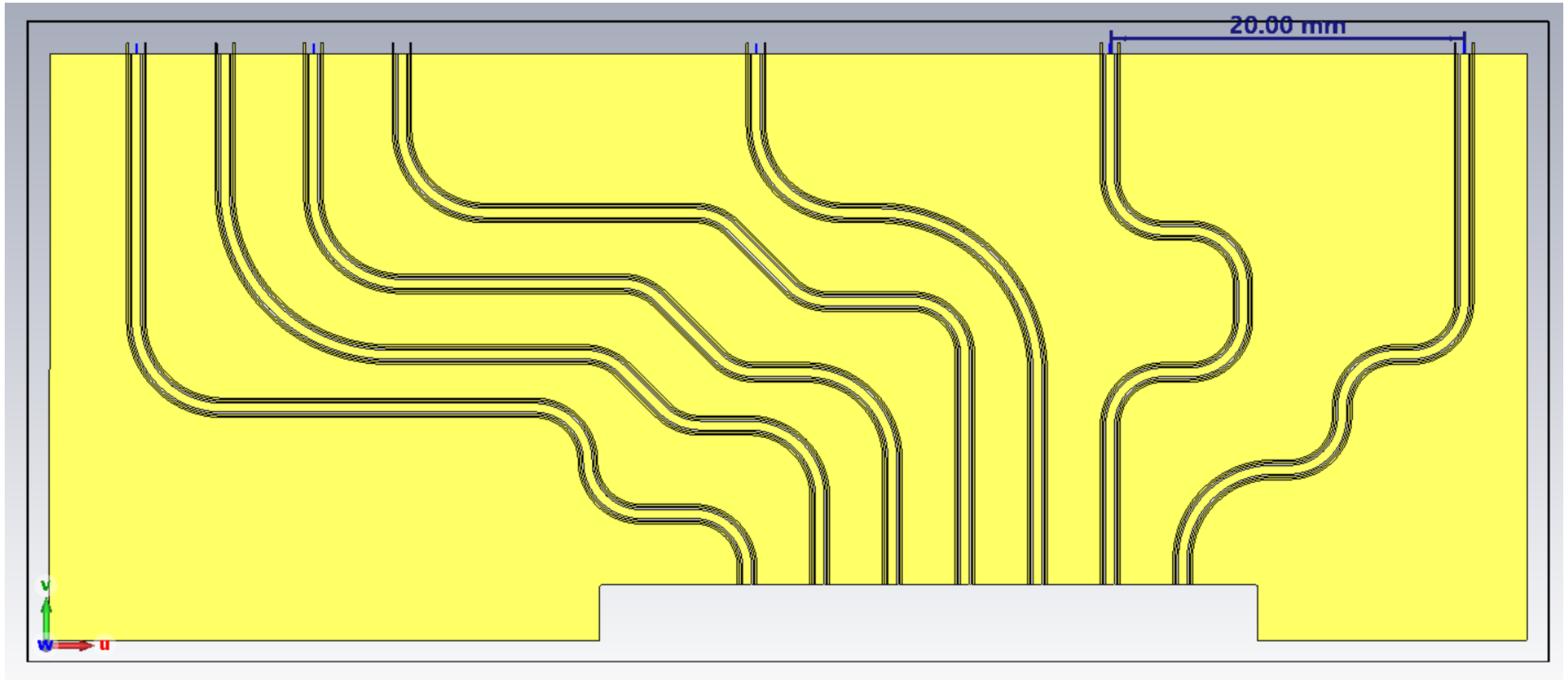
转接固定器

转接器设计要求——等相位线

设计方案：CST设计等长线（不用严格等长，可以调节），转角不能急转，弧度要足够大。

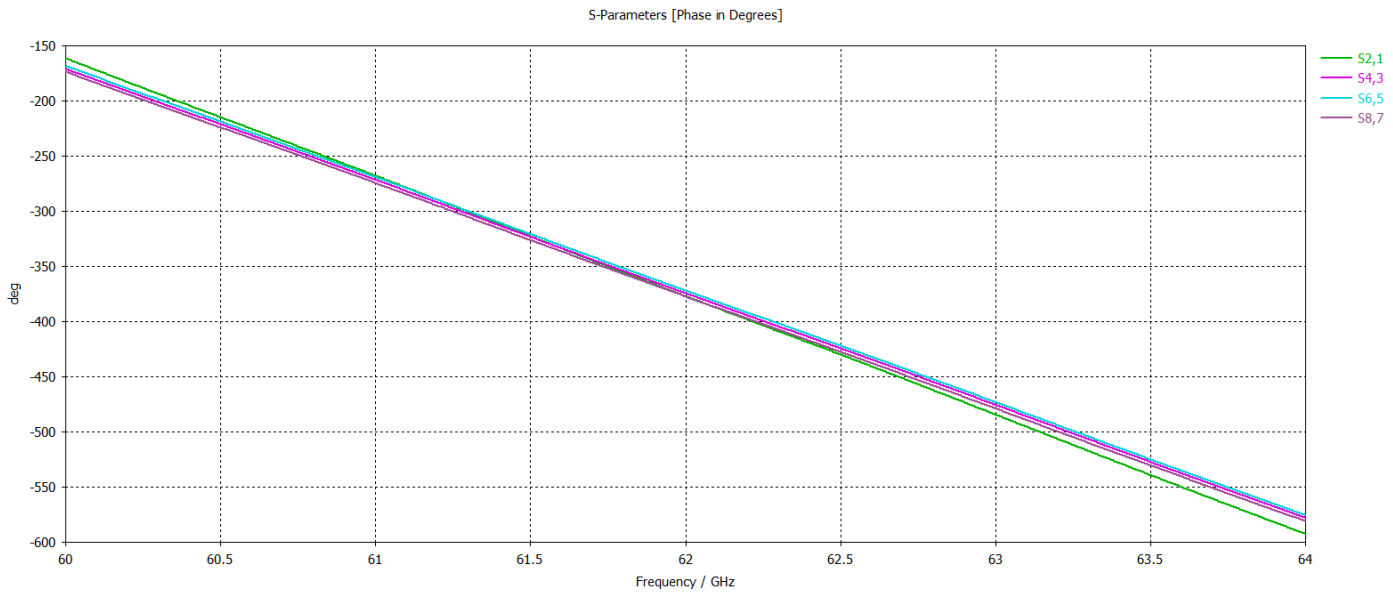


设计方案：CST设计利用画出的线切割出GCPW的辐射缝隙和电壁并进行仿真。

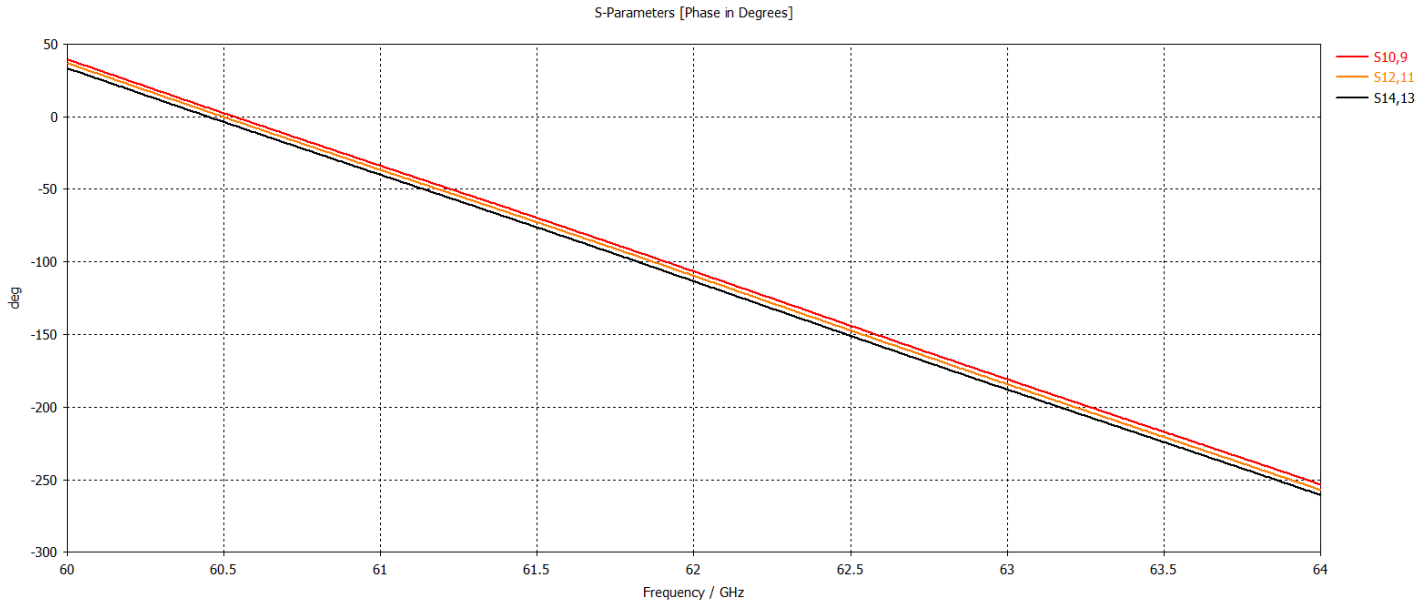


"E:\课题组项目\天线设计\波束扫描天线设计项目\转接器\转接器 - last.cst"

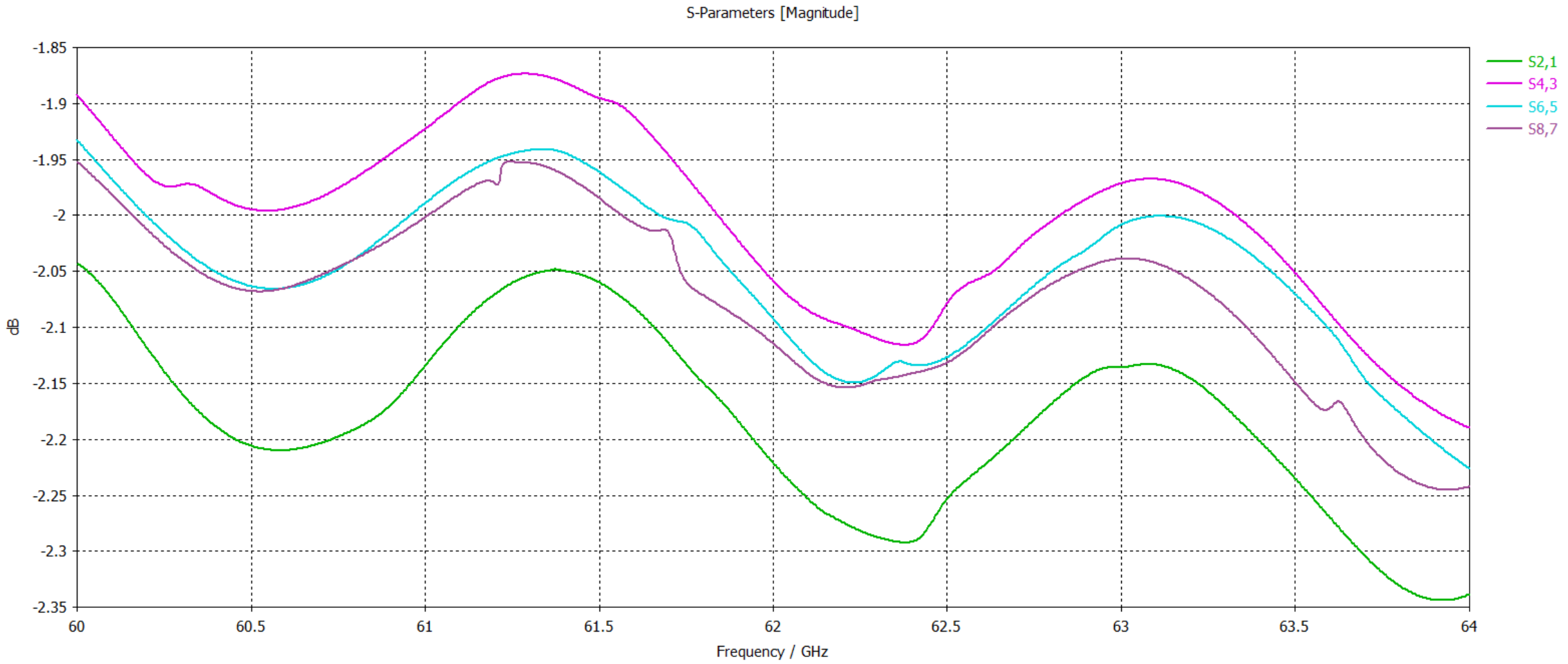
接收天线通道



发射天线通道

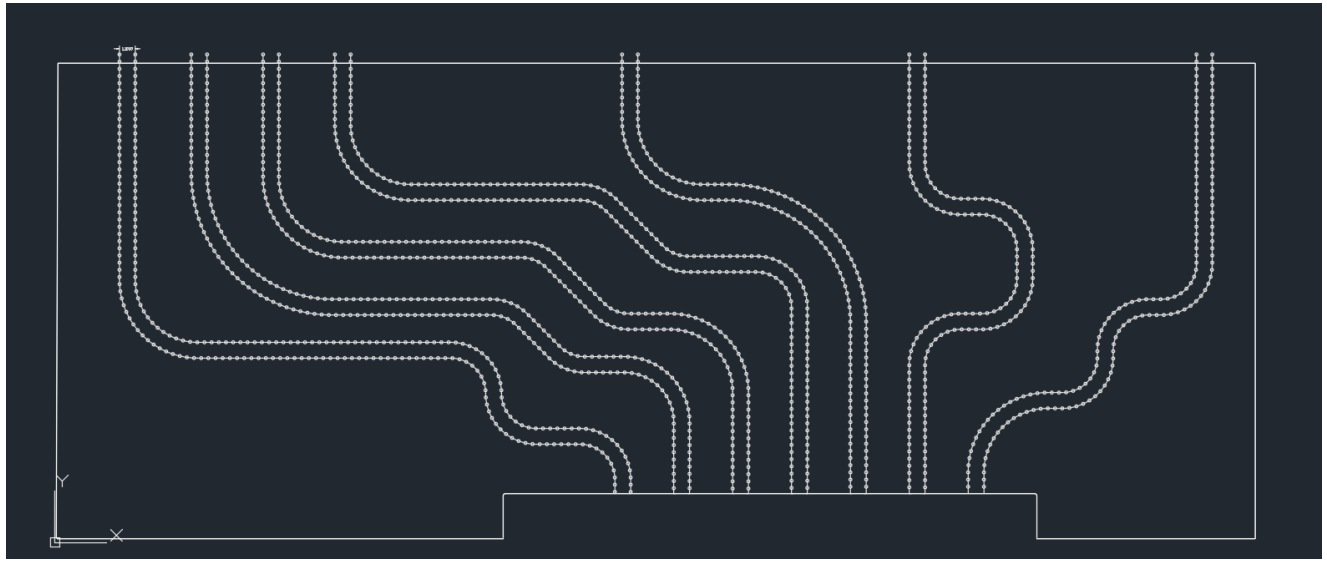


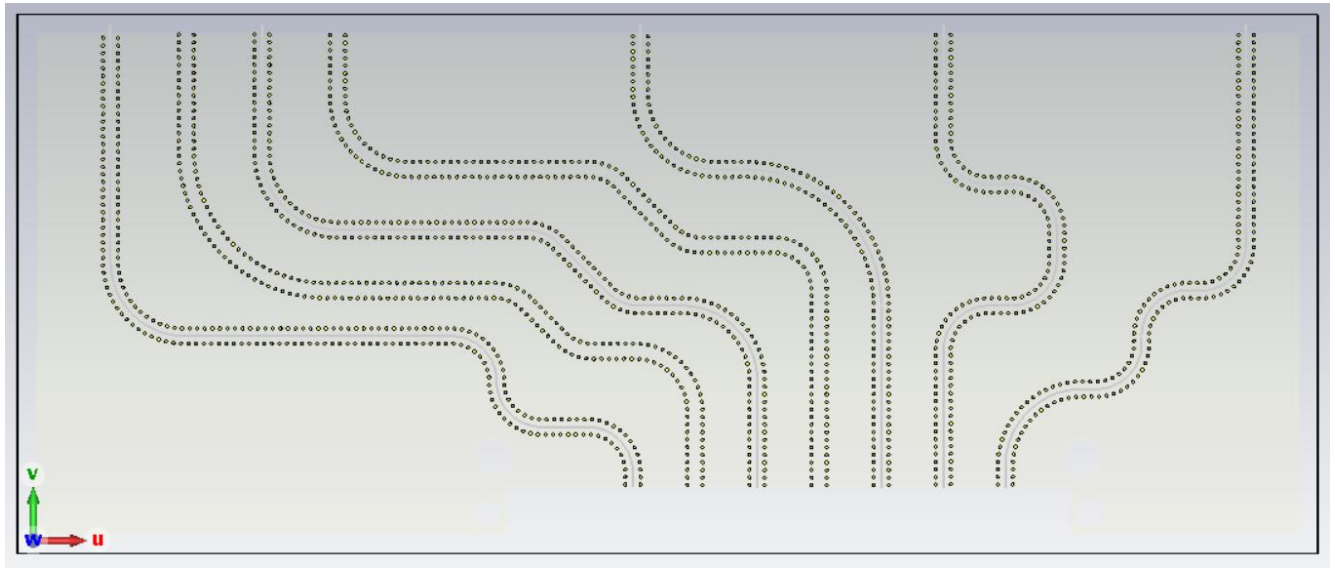
能量会不可避免的有一定的损耗。



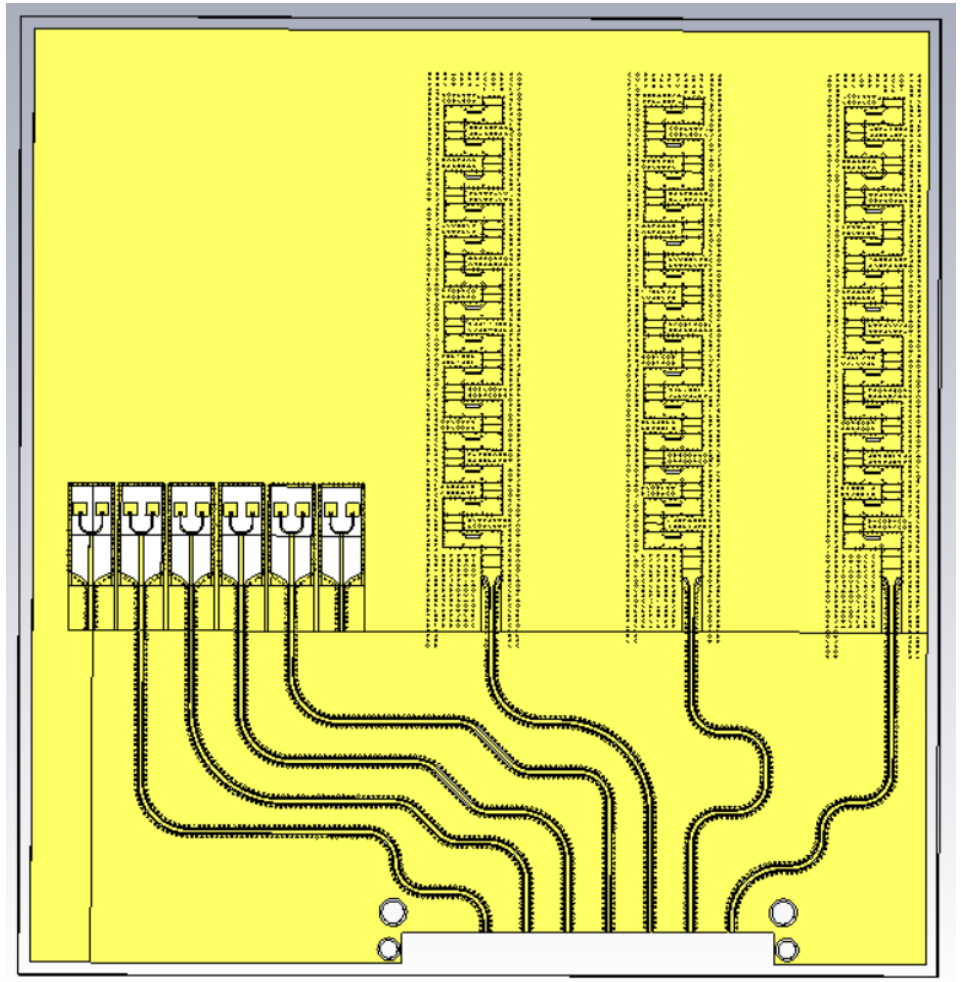
电壁转为SIW（过孔） 导出平面图，利用CAD进行路径式阵列画圆圈，导回到CST将画出的圆拉起来变为过孔。

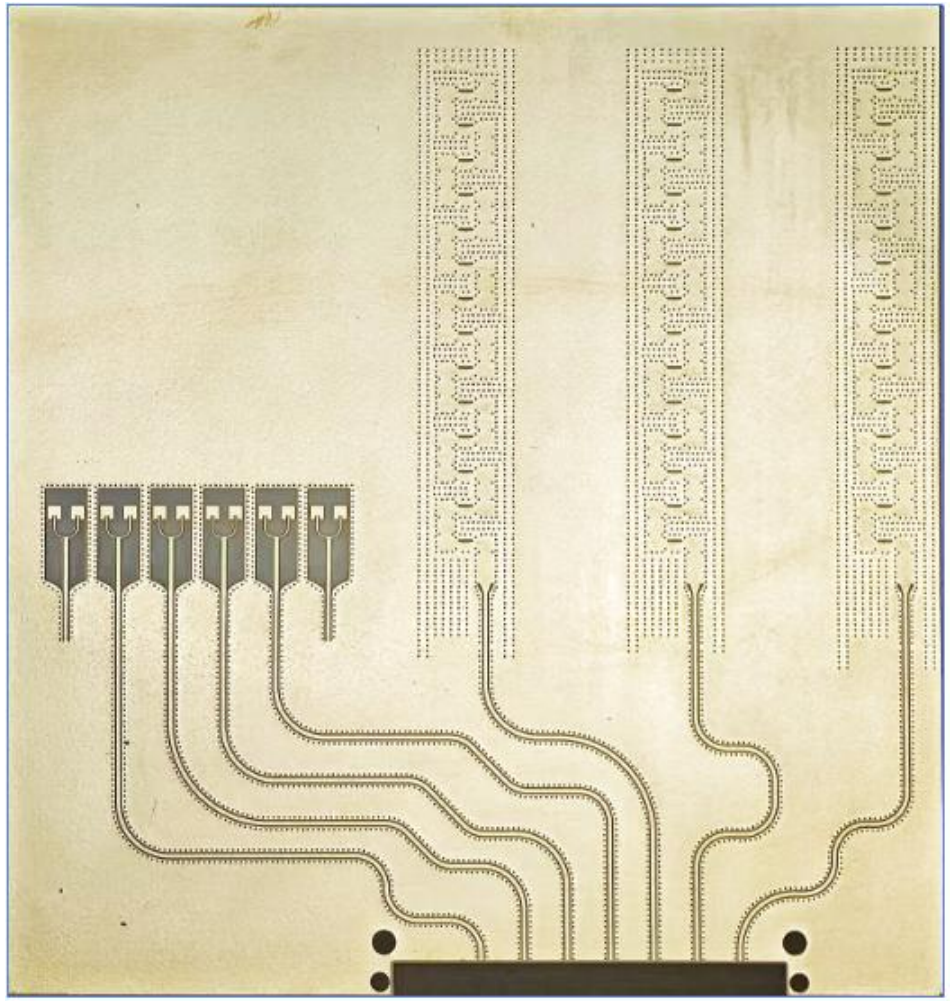
E:\课题组项目\天线设计\CAD





整体结构与加工图片（三发四收）





总结归纳

遇到过的问题：

1.发射天线设计时谐振腔频率调节不当：

（1）特征本征模确定腔体大致尺寸。

（2）将耦合缝隙削弱确定各个腔体频率是否正确，进行尺寸微调。

（3）利用耦合矩阵调整耦合强度（连接尺寸），使独立的谐振频率连接。

（4）微调辐射缝隙和输入输出端口。

2.电壁转SIW出现错误：

依照理论结算基片集成波导的尺寸变化为 $W_{\mathrm{siw}} = W + 2 * (d + s)$,其中W为电壁尺寸，d为金属孔径，s为孔间距。但是仿真结果与计算结果有所偏差，因此需要对部分孔位置进行微调。

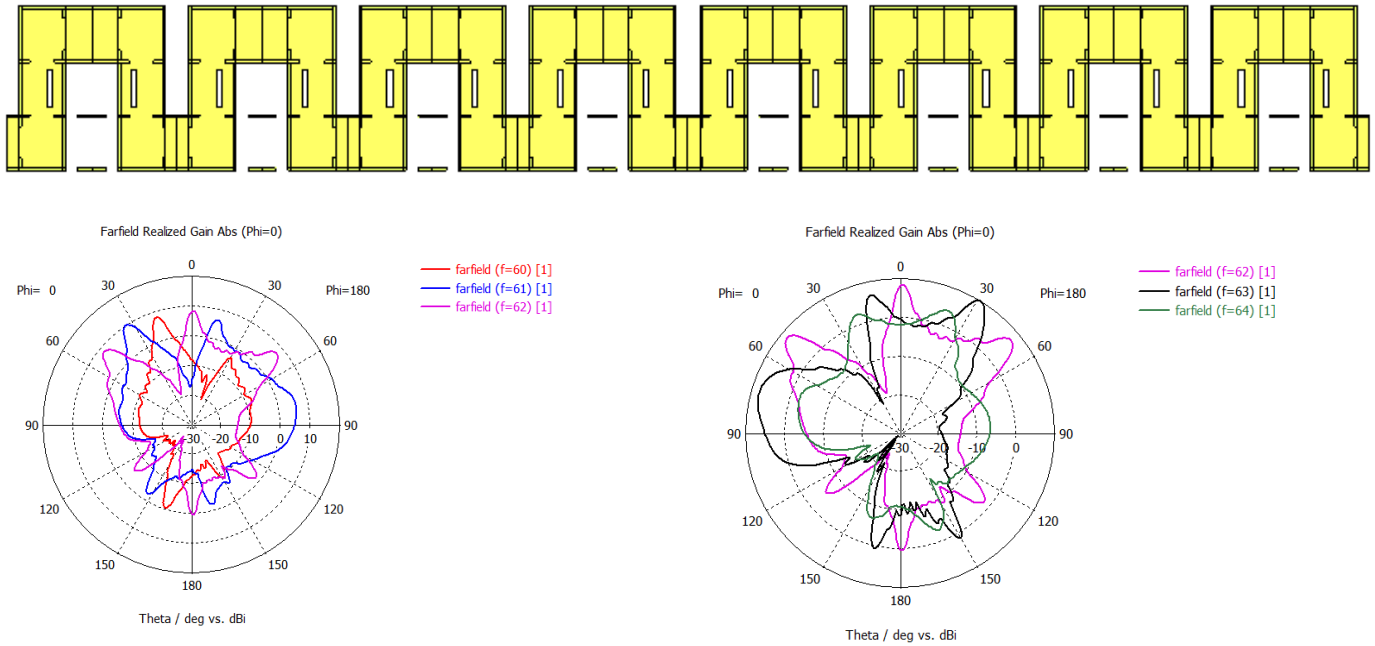
总结归纳

遇到过的问题：

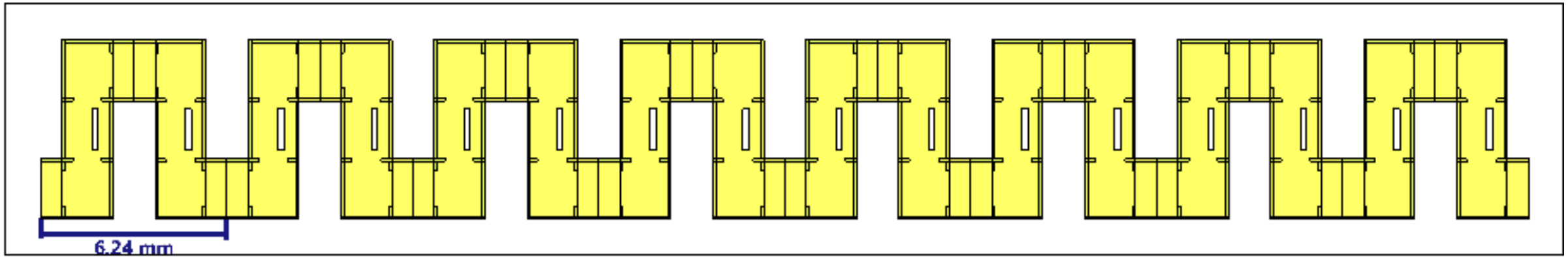
3.波束出现大偏移：缝隙位置调整出错，原设计如下图所示，扇形对称并且保证缝隙距离相同，但远场波束出现严重旁瓣影响。

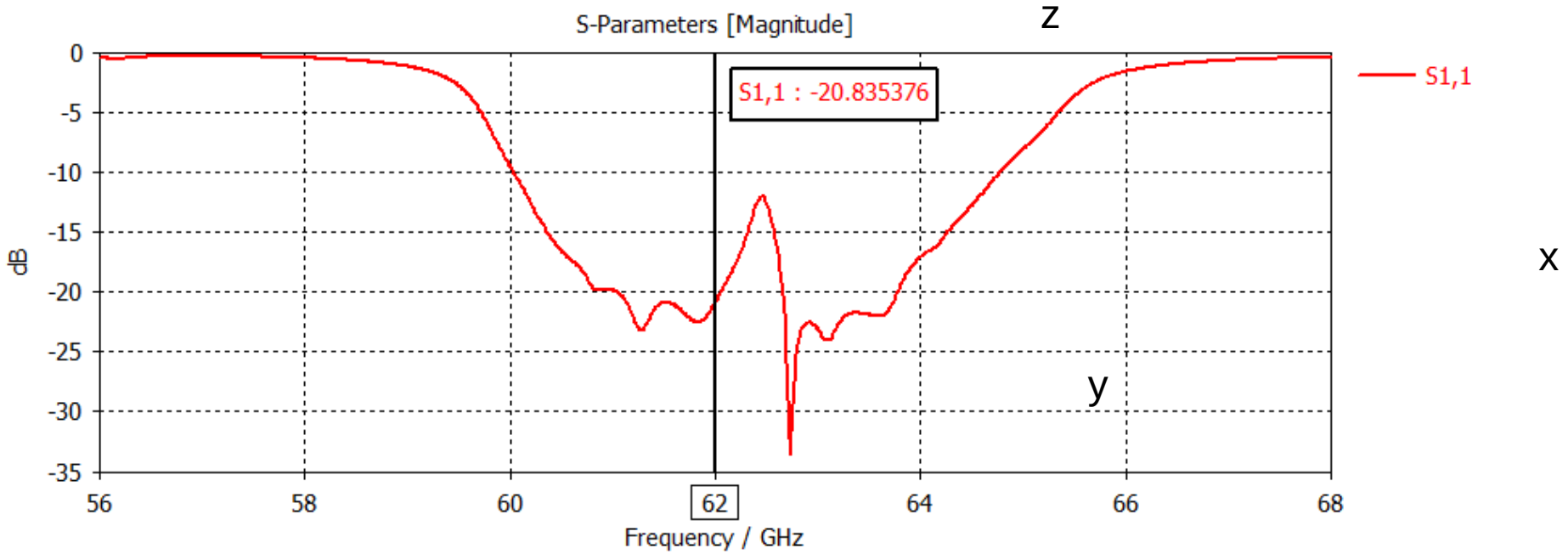
分析：出现严重旁瓣说明不是所有缝隙的辐射方向都一致，产生了两种独立的辐射方向，需要对对称的缝隙位置进行调整

解决方法：腔体调整为完全折叠对称，辐射缝隙位置翻转，结果辐射方向达到预期



遇到过的问题：
3.波束出现大偏移：优化结果图



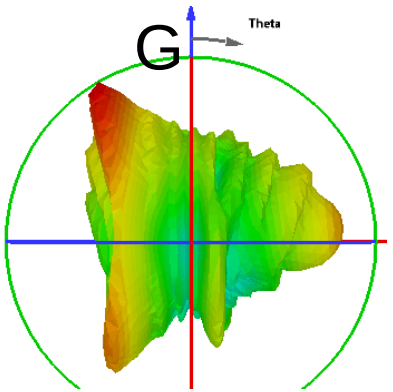


z

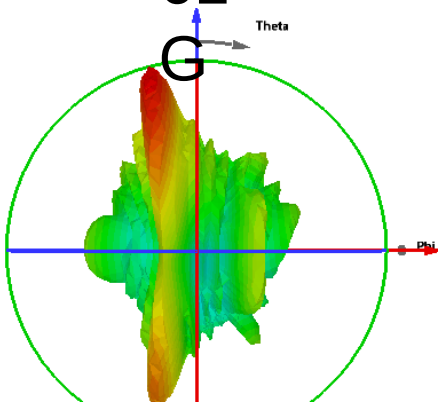
x

y

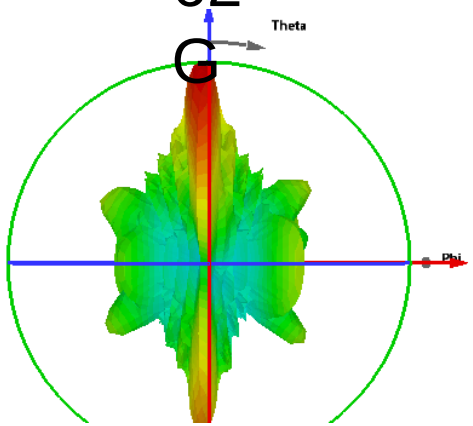
60 G



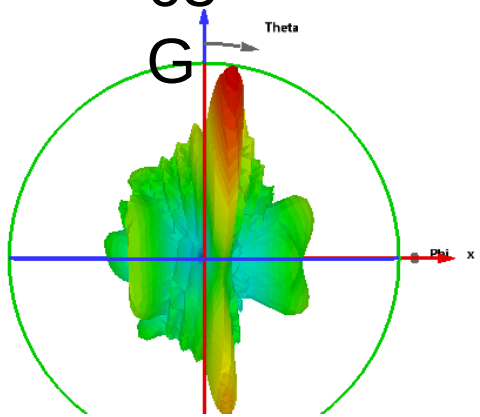
61 G



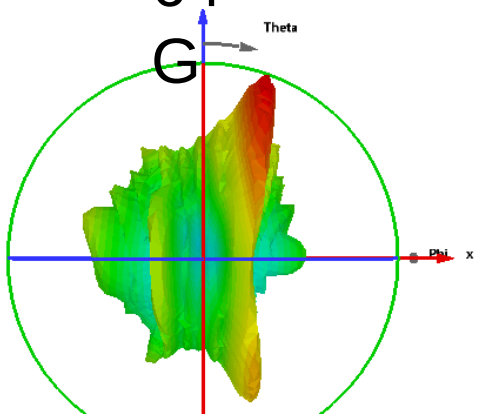
62 G



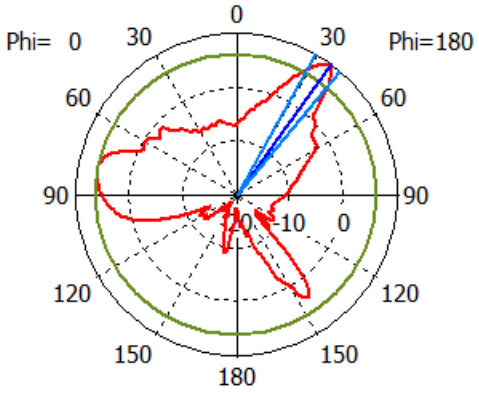
63 G

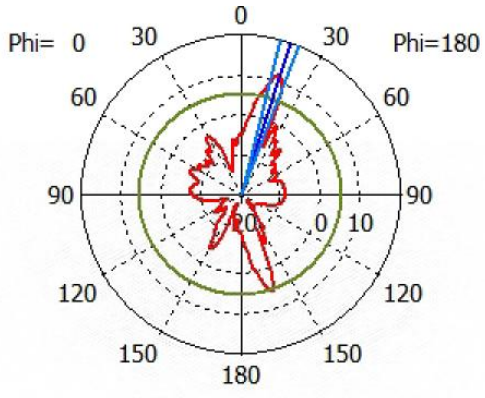


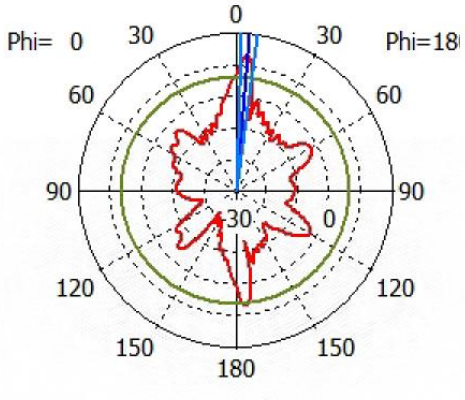
64 G

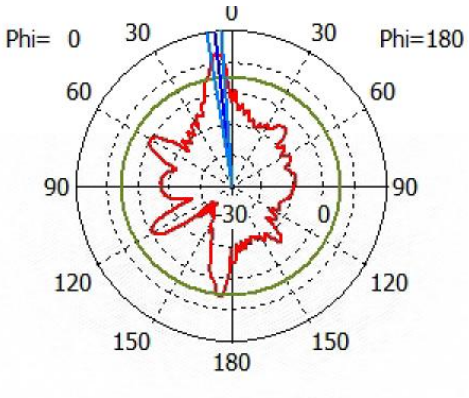


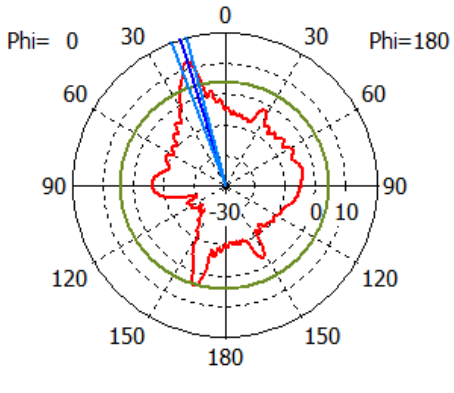
Farfield Realized Gain Abs (Phi=0)











36度
9.56dB
;

18度
11.5dB
;

5度
12.8dB
;

-6度
12.8dB
;

-17度
12.5dB
;

总结归纳

遇到过的问题：

4.中心频率波束未对准0度：调整波导的长度，进而调整波束的相位偏移

5.OSB开放阻带：在大多数漏波天线中，由于结构设计的原因，在某些频率范围内，天线无法在宽边（broadside）方向进行有效辐射，这个频率范围被称为开放阻带。在OSB频段内，天线的辐射效率显著降低，导致增益下降，无法实现预期的辐射性能。

解决方法：对辐射缝隙开一条小的横向缝隙，不影响辐射的情况下破除天线的周期性，降低OSB的影响

6.GCPW转波导设计：由于转接电路板的需要，需将天线的波导转为GCPW，设计时试图使用阻抗匹配法（四分之一波长阻抗变换器）设计中间的连接部位，发现仿真后效果并不好，最后使用上述直连方法完成设计。

四分之一波长阻抗变换器：通过在源和负载之间插入一段特性阻抗为特定值的传输线，可以实现阻抗匹配。这段传输线的长度为四分之一波长，特性阻抗 $Z0$ 通常为源阻抗 Zs 和负载阻抗 ZL 的几何平均值： Zs为55欧姆 ZL为105欧姆

$$Z_0 = \sqrt{Z_s \cdot Z_L}$$

总结归纳

遇到过的问题：

7.接收天线设计：使用定波束天线，保证天线的波束在扫描角度内可以固定接收信号。尝试过几种不同的天线结构，最后选择最常用的patch天线，利用并列模式压缩波束角度使增益较强，能量更为集中

8.转接器设计电壁转为SIW：在CST中无法直接进行如此复杂的通孔排列，电壁不能直接用绘制的等相位线进行偏印处理。最终决定借助CAD画图软件将通孔进行线路阵列化后导入CST完成SIW设计。

总结归纳

遇到过的问题：

9.设计的天线无法加工

（1）介质板厚度：在设计时没有完全了解盲目设计了0.2mm的尺寸，但加工板材尺寸为0.254mm导致设计需进行返工调整。

（2）孔径和孔距：在设计时使用0.1mm孔径和0.2mm孔间距，加工时发现加工商的打孔器具为0.2mm孔径，孔间距为0.5mm，导致无法加工，需要重新设计调整。

（3）金属层厚：完成上述调整后发现，原定为0.5oz铜进行沉金，作为表面金属层处理工艺，但由于通孔也需要镀铜，所以改为1oz，导致表面厚度翻倍，设计需要返工

（4）加工精度：由于工作频段在60GHz以上，天线尺寸较小，精度较高，完成加工后用需用显微镜进行尺寸测量，发现有些尺寸的精度没有达到合同要求，尤其是辐射缝隙的尺寸差别较大，重新要求加工厂进行加工。

总结经验：在设计之前一定要完全弄清楚可加工的条件以及加工组件的各项尺寸和精度，不能脱离生产盲目设计，否则会出现无法加工的情况。天线设计精度需要调低，适应性强的天线更适合生产和加工使用。

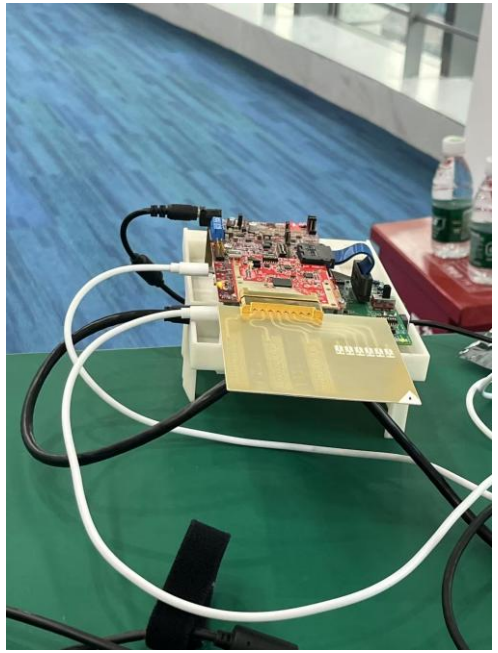
总结归纳

未来需解决问题：

1.损耗过大：

由于工作频率过高（60GHz），导线、转接器、GCPW等损耗过大，导致辐射出的能量较小。

与解决方法：完成板基天线设计，减少天线接上测试版时的损耗，GCPW上加反射板，防止能量的辐射。



总结归纳

未来需解决问题：

2.扫描角度不够大

需要新的天线结构，或者进行四阶谐振腔的天线设计

3.天线精度要求过高，调参过于灵敏，需要对各项参数进行优化，使其误差范围可以达到0.1mm以上，方便后续的应用和生产。

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